Stochastic Reconfiguration as Statistical Filtering for Overparameterized Neural Quantum States

Tak Hur
Department of Statistics and Data Science
Yonsei University
Seoul, Republic of Korea
takh0404@yonsei.ac.kr

Abstract

Stochastic reconfiguration (SR) is the standard optimizer for neural quantum states (NQS), but modern NQS often have far more parameters than Monte Carlo samples. We show that in this regime the diagonal shift is more than a numerical stabilizer. It acts as a statistical filter for finite-sample generalization. At a fixed wave function, SR is ridge regression from tangent features to the centered local energy. Its residual is the *expressivity gap*, the part of imaginary-time evolution outside the current tangent space. This gap is orthogonal to the tangent space in population, but finite batches make it act as noise that SR can overfit. The shift therefore balances shrinkage of useful update directions against variance from fitting sampled residuals. Exact 4×4 J_1-J_2 experiments separate two effects of overparameterization. Larger tangent spaces help when they reduce the expressivity gap, but they can hurt when they overfit a fixed gap. In a 100-site transverse-field Ising family trained with a foundation NQS, validation risk is U-shaped in the shift while variance decreases, matching the noisy-ridge model. This view leads to multi-shift SR (MS-SR), which averages independent ridge solves at several shifts to form a richer, lower-variance spectral filter. Checkpoint diagnostics show that its benefit is regime-dependent, and controlled late-stage branches compare MS-SR with SR and SPRING at equal solve budget. MS-SR closely tracks SR and finishes below SPRING in both branches, while the same diagnostics identify regimes where a single oracle-tuned shift is preferable.

1 Introduction

Stochastic Reconfiguration (SR) remains the dominant optimizer for neural quantum states (NQS), but the regime in which it operates has changed fundamentally. Classical variational Monte Carlo with Jastrow, Gutzwiller, and projected mean-field ansatzes, as well as early RBM-based NQS, had modest parameter counts [14, 31, 1, 28, 24, 25, 2]. Kernel-form SR (minSR) removed the need to form the full $P \times P$ quantum geometric tensor (QGT), enabling the shift to transformer-based and foundation-style wave functions with millions of parameters [4, 9, 17, 10, 30, 3, 19, 26, 12, 16, 18]. However, minSR still solves a dense $N_s \times N_s$ system built from the empirical NTK matrix, at cost $O(N_s^2 P)$ to form and $O(N_s^3)$ to solve. Increasing N_s therefore remains the practical bottleneck, pushing modern NQS from the sample-rich regime of classical VMC ($P < N_s$) into a parameter-rich regime where P exceeds N_s by one to three orders of magnitude (Figure 1).

Each SR update solves a linear system with an empirical QGT, which can be ill-conditioned when estimated from finitely many Monte Carlo samples. The standard remedy is to add a diagonal shift before solving [24, 25, 2]. Historically, this shift was therefore understood mainly as a numerical conditioning device. In the modern minSR regime, however, the same operation is applied when the

40th Conference on Neural Information Processing Systems (NeurIPS 2026).

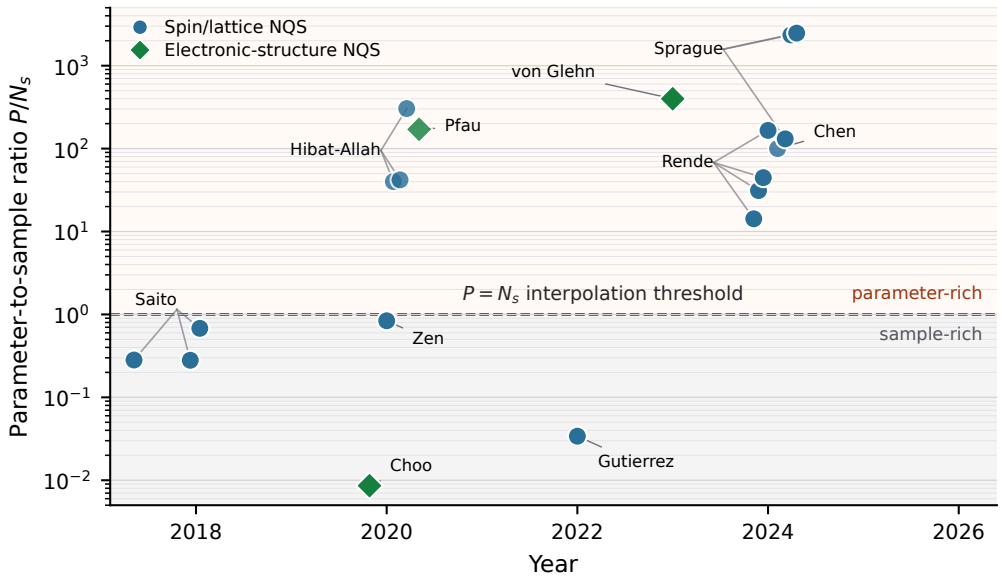


Figure 1: Historical shift in the parameter-to-sample ratio used by VMC/NQS calculations. The dashed line marks $P = N_s$. Marker shape denotes domain: circles are spin/lattice NQS and diamonds are electronic-structure NQS. Values use each source's parameter-count convention and raw optimizer batch size; approximate source-reported counts are included as order-of-magnitude points. Sources: [20, 21, 4, 32, 10, 17, 8, 30, 19, 3, 26].

number of variational parameters can far exceed the number of samples, so its role may be statistical as well as numerical. This raises the central question of this paper: is the diagonal shift in SR merely a numerical stabilizer, or is it a statistical filter that makes overparameterized finite-sample updates generalize, and can this view lead to better filters than a single λ?

This question exposes the central tension of modern NQS optimization. More expressive neural quantum states can reduce the *expressivity gap*, the component of imaginary-time evolution outside the current tangent space. However, finite-sample SR can also overfit the part of this gap that remains. The central contributions of this work are:

- We formulate SR as a misspecified ridge-regression problem. The expressivity gap acts as finite-sample residual noise, yielding an exact held-out error identity and a spectral risk approximation for the bias–variance role of the diagonal shift λ.
- We test this mechanism in both exactly tractable small-scale and modern large-scale NQS settings. Exact 4×4 matched-noise and matched-state experiments separate finite-sample overfitting from expressivity-gap reduction. Large-scale shared-checkpoint diagnostics show the predicted U-shaped validation risk and monotone variance reduction as λ increases.
- We use the filter view to design multi-shift SR (MS-SR), which combines independent ridge solves at multiple shifts. Checkpoint-local ablations isolate multi-shift filtering, independent-batch averaging, and learned stacking. In two equal-budget late-stage branches, fixed-grid MS-SR closely tracks SR and finishes below SPRING; robustness and wall-clock studies delimit when the additional batches are worthwhile.

Section 2 develops the noisy ridge-regression view of SR. Sections 3 and 4 validate its predictions in exact small-scale and modern large-scale settings. Section 5 shows that the same perspective leads naturally to MS-SR as a stronger practical alternative to single-shift SR.

2 SR as Misspecified Ridge Regression

Throughout this section we fix a checkpoint θ and assume that the wavefunction and Hamiltonian are real in the sampling basis for notational simplicity. The complex case is obtained by replacing transposes with Hermitian conjugates (see Appendix B for detail).

SR as tangent-space regression. Let ψ_θ be a neural quantum state with Born distribution $\pi(x) = |\psi_\theta(x)|^2/\|\psi_\theta\|^2$. Define the centered tangent features and centered local energy by

$$O_c(x) = \nabla_\theta \log \psi_\theta(x) - \mathbb{E}_\pi[\nabla_\theta \log \psi_\theta], \qquad H_{\mathrm{loc},c}(x) = H_{\mathrm{loc}}(x) - E(\theta),$$

where $H_{\mathrm{loc}}(x) = \langle x|H|\psi_\theta\rangle/\langle x|\psi_\theta\rangle$ and $E(\theta) = \mathbb{E}_\pi[H_{\mathrm{loc}}]$. Under normalized imaginary-time evolution, the log amplitude changes as $\Delta \log \psi_\theta(x) = -\eta H_{\mathrm{loc},c}(x) + O(\eta^2)$. A parameter update $\theta^+ = \theta - \eta\delta$ changes the log amplitude as $\log \psi_{\theta^+}(x) - \log \psi_\theta(x) = -\eta O_c(x)^T\delta + O(\eta^2)$, up to an x-independent normalization term. Matching these first-order changes shows that fixed-checkpoint SR fits tangent predictions $O_c(x)^T\delta$ to the centered local energy $H_{\mathrm{loc},c}(x)$.

Writing $S^\dagger$ for the Moore–Penrose pseudoinverse of $S = \mathbb{E}_\pi[O_c(x)O_c(x)^T]$, and setting $g = \mathbb{E}_\pi[O_c(x)H_{\mathrm{loc},c}(x)]$, the ideal fixed-checkpoint SR direction is the population least-squares solution

$$\delta^* = S^\dagger g \in \arg\min_\delta \mathbb{E}_\pi\Big[\big(O_c(x)^T\delta - H_{\mathrm{loc},c}(x)\big)^2\Big]. \tag{1}$$

Empirical SR as misspecified ridge regression. The centered local energy need not be exactly representable by the current tangent features. We define the tangent-space expressivity gap $\epsilon(x) = H_{\mathrm{loc},c}(x) - O_c(x)^T\delta^*$. The normal equations imply $\mathbb{E}_\pi[\epsilon(x)O_c(x)] = 0$, and the gap variance is $\sigma^2_{\mathrm{gap}} = \mathbb{E}_\pi[\epsilon(x)^2] = \mathrm{Var}_\pi(H_{\mathrm{loc}}) - g^T S^\dagger g$. We call $\epsilon(x)$ the misspecification noise: it is the part of the imaginary-time target that lies outside the current tangent space.

Given a Monte Carlo batch $\mathcal{D} = \{x_j\}_{j=1}^{N_s}$, empirical SR solves the ridge problem

$$\hat\delta_\lambda = (\hat S + \lambda I)^{-1}\hat g, \qquad \hat S = \frac{1}{N_s}\sum_{j=1}^{N_s} O_c(x_j)O_c(x_j)^T, \qquad \hat g = \frac{1}{N_s}\sum_{j=1}^{N_s} O_c(x_j)H_{\mathrm{loc},c}(x_j).$$

Although $\epsilon(x)$ is orthogonal to the tangent features in population, it is still present in the sampled targets $H_{\mathrm{loc},c}(x_j)$. An overparameterized empirical solve can therefore memorize this residual on the batch, even though it does not correspond to a generalizable tangent-space update. In this sense, the expressivity gap behaves as regression noise.

Excess risk analysis. Here, we are interested in S-norm excess risk $\mathcal{E}_\lambda = \mathbb{E}_\mathcal{D}\|\hat\delta_\lambda - \delta^*\|_S^2$, where the expectation is over the Monte Carlo batch used to form $\hat\delta_\lambda$. This is the out-of-sample error of the checkpoint-local ridge problem with features $O_c(x)$, target $H_{\mathrm{loc},c}(x)$, and residual noise $\epsilon(x)$.

The exact bias–variance identity, together with the spectral proxy (see Appendix B for details), gives

$$\mathcal{E}_\lambda = \underbrace{\Big\|\mathbb{E}_\mathcal{D}[\hat\delta_\lambda] - \delta^*\Big\|_S^2}_{\text{Bias}} + \underbrace{\mathbb{E}_\mathcal{D}\Big\|\hat\delta_\lambda - \mathbb{E}_\mathcal{D}[\hat\delta_\lambda]\Big\|_S^2}_{\text{Variance}} \approx \underbrace{\sum_i s_i\left(\frac{\lambda}{s_i+\lambda}\right)^2(\beta_i^*)^2}_{\mathcal{B}_{\mathrm{shrink}}(\lambda)} + \underbrace{\frac{\sigma^2_{\mathrm{gap}}}{N_s}\sum_i\left(\frac{s_i}{s_i+\lambda}\right)^2}_{\mathcal{V}_{\mathrm{gap}}(\lambda)}, \tag{2}$$

where $S = V\,\mathrm{diag}(s_i)V^T$ and $\beta_i^* = (V^T\delta^*)_i$. The first term is shrinkage bias: it grows when λ damps useful components of δ^*. The second term is finite-sample variance: it decreases because the same filter suppresses fits to sampled gap residuals. Thus λ is a statistical regularizer, not only a numerical tolerance. Standard SR, however, is restricted to the one-parameter filter family $s/(s+\lambda)$, motivating the multi-λ filters in Section 5.

This S-norm excess risk is physically relevant because it is closely related to the infidelity, a standard distance measure between quantum states. For bounded updates,

$$\mathcal{I}_\theta(\delta,\delta') = 1 - |\langle\psi_{\theta-\eta\delta}|\psi_{\theta-\eta\delta'}\rangle|^2 = \eta^2\,\|\delta-\delta'\|_S^2 + O(\eta^3). \tag{3}$$

Therefore $\mathcal{E}_\lambda$ controls the expected local infidelity between the ideal population update and the empirical SR update, up to the factor η^2 and higher-order terms.

Large-system diagnostics for excess risk and variance. For small systems, the population quantities S, δ^*, σ^2_{gap}, and $\|\hat{\delta}_\lambda - \delta^*\|^2_S$ can be computed by exact diagonalization. This is impossible in the large scale regimes where modern NQS are tuned and compared. We therefore use two fixed-checkpoint diagnostics that require only held-out predictions.

The first diagnostic is a validation residual. Let $\mathcal{D}_{\mathrm{val}}$ be independent of the batches used to compute the SR updates. Since $H_{\mathrm{loc},c}(x) = O_c(x)^T\delta^* + \epsilon(x)$ with $\mathbb{E}_\pi[\epsilon(x)O_c(x)] = 0$, any fixed update δ satisfies

$$R_{\mathrm{val}}(\delta) = \frac{1}{|\mathcal{D}_{\mathrm{val}}|}\sum_{x\in\mathcal{D}_{\mathrm{val}}}\left(O_c(x)^T\delta - H_{\mathrm{loc},c}(x)\right)^2, \qquad \mathbb{E}_{\mathcal{D}_{\mathrm{val}}}\left[R_{\mathrm{val}}(\delta)\right] = \|\delta - \delta^*\|^2_S + \sigma^2_{\mathrm{gap}}. \quad (4)$$

Thus, for independent SR updates $\hat{\delta}_{\lambda,1},\ldots,\hat{\delta}_{\lambda,m}$ at the same checkpoint, we use $\widehat{\mathcal{E}}_{\mathrm{val}}(\lambda) = m^{-1}\sum_{j=1}^m R_{\mathrm{val}}(\hat{\delta}_{\lambda,j})$. Its expectation is $\mathcal{E}_\lambda + \sigma^2_{\mathrm{gap}}$, so it tracks excess risk plus the expressivity-gap.

The second diagnostic measures how much the update varies across independent training batches. With $\bar{\delta}_\lambda = m^{-1}\sum_j \hat{\delta}_{\lambda,j}$, define

$$\mathcal{V}_{\mathrm{mb}}(\lambda) = \frac{1}{m-1}\sum_{j=1}^m \frac{1}{|\mathcal{D}_{\mathrm{val}}|}\sum_{x\in\mathcal{D}_{\mathrm{val}}}\left(O_c(x)^T(\hat{\delta}_{\lambda,j} - \bar{\delta}_\lambda)\right)^2. \quad (5)$$

As m and the validation batch grow, this converges to the variance component of the excess risk. Both diagnostics are practical at large scale because they only require held-out tangent-feature predictions. Since they do not require forming or solving a validation QGT or NTK system, they can be evaluated on validation batches much larger than the SR training batch with modest computational cost.

Together, these diagnostics test the bias and variance role of the diagonal shift at a fixed checkpoint. Increasing λ first suppresses noisy small-eigenvalue directions, so $\widehat{\mathcal{E}}_{\mathrm{val}}$ can decrease as variance falls. Once λ also suppresses useful update directions, shrinkage bias dominates and the validation curve rises. The predicted signature is therefore a U-shaped $\widehat{\mathcal{E}}_{\mathrm{val}}$ with a monotonically decreasing $\mathcal{V}_{\mathrm{mb}}$.

3 Small-scale Dissection of Expressivity and Overfitting in SR

This section tests the mechanism of Section 2 in an exactly diagonalizable spin-$1/2$ system. We use the periodic 4×4 square-lattice J_1-J_2 Heisenberg model $H = J_1\sum_{\langle i,j\rangle}\mathbf{S}_i\cdot\mathbf{S}_j + J_2\sum_{\langle\langle i,j\rangle\rangle}\mathbf{S}_i\cdot\mathbf{S}_j$. We set $J_2/J_1 = 0.5$, which is a strongly frustrated point. Here $\mathbf{S}_i$ is the spin-$1/2$ operator at site i, while $\langle i,j\rangle$ and $\langle\langle i,j\rangle\rangle$ denote nearest-neighbor and next-nearest-neighbor bonds.

All SR solves use a fixed sample budget of $N_s = 4096$. We compare an underparameterized RBM with $P/N_s = 0.27$ to an overparameterized Vision Transformer with $P/N_s = 3.28$. For the selected comparisons, we draw 100 independent training batches for matched noise and 10 for matched state, then compute the corresponding empirical shifted-SR updates $\hat{\delta}_\lambda(\mathcal{D}_b)$. Because the 4×4 system is exactly diagonalizable, the population QGT S, the population target update δ^*, and the expressivity-gap variance σ^2_{gap} are computed exactly over the full Hilbert space. We estimate the excess risk by averaging $\|\hat{\delta}_\lambda(\mathcal{D}_b) - \delta^*\|^2_S$ over the independent batches, and report the sample mean with one sample standard error (the trial sample standard deviation divided by the square root of the number of batches). Thus, unlike the large-system diagnostic $\widehat{\mathcal{E}}_{\mathrm{val}}$, this small-scale measurement uses the exact population geometry rather than a validation approximation. We use the near-ridgeless shift $\lambda = 10^{-9}$, which deliberately exposes the tendency of empirical SR to fit sample-specific expressivity-gap noise. Equivalently, Figure 2 visualizes this excess risk pointwise: deviations from the diagonal measure how much the empirical prediction $O_c(x)^T\hat{\delta}_\lambda$ differs from the ideal tangent-space prediction $O_c(x)^T\delta^*$ on training (red) and held-out (blue) configurations.

The experiment separates two effects that are coupled. The matched-noise comparison keeps σ^2_{gap} approximately fixed and tests the finite-sample cost of a larger tangent space. The matched-state comparison keeps the represented wavefunction approximately fixed and tests the expressivity benefit of the larger tangent space. Full model and optimization details are given in Appendix C.

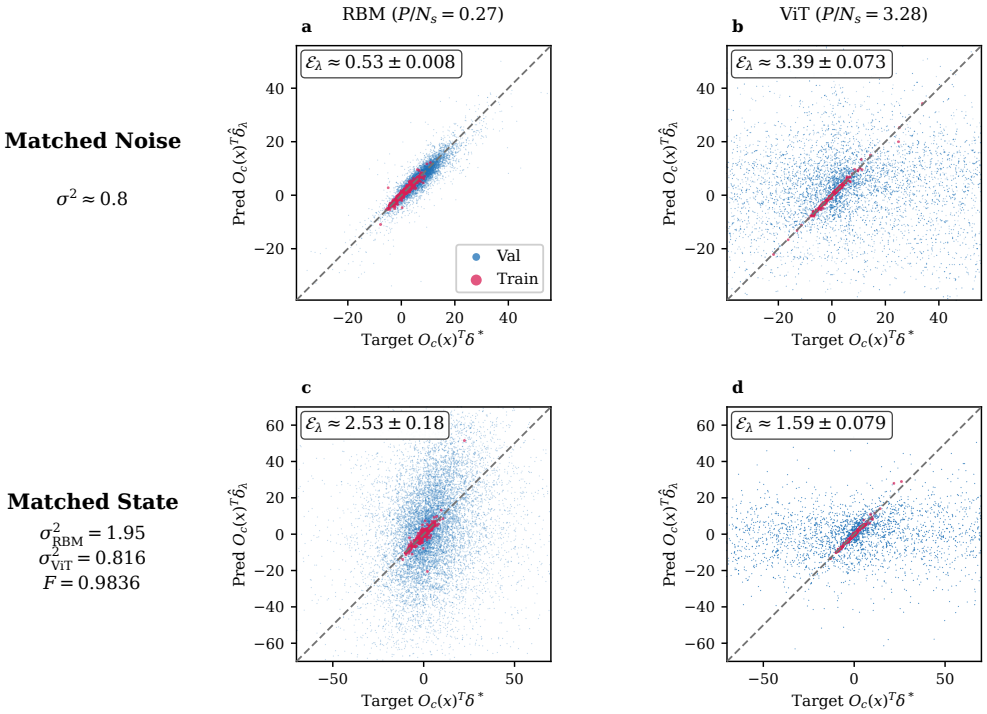


Figure 2: Small-scale SR risk geometry for the periodic 4×4 J_1-J_2 model at $N_s = 4096$ and $\lambda = 10^{-9}$. Panels compare empirical predictions (y-axis) with ideal tangent-space predictions (x-axis). Matched-noise exposes ViT overfitting at comparable noise σ^2_{gap}. Matched-state shows the ViT's larger tangent space reduces the expressivity gap and excess risk. Reported uncertainties are one sample standard error across the 100 matched-noise or 10 matched-state independently resampled training batches.

Matched-noise comparison. To test the finite-sample cost of overparameterization, we select independently trained RBM and ViT checkpoints with comparable expressivity-gap variance. In Figure 2 panel (a) and (b), both values of σ^2_{gap} are close to 0.8. At $\lambda = 10^{-9}$, the RBM remains close to the ideal population update, with $\mathcal{E}_\lambda \approx 0.53 \pm 0.008$. The ViT has comparable misspecification noise, but its empirical solve fits the sampled residual much more strongly, increasing the excess-risk estimate to $\mathcal{E}_\lambda \approx 3.39 \pm 0.073$. This is the variance-driven failure mode predicted by Eq. (2). When the expressivity gap is held fixed, a larger tangent space gives the empirical SR update more directions in which to fit sample-specific noise.

Matched-state comparison. To isolate the benefit of a larger tangent space, we freeze a partially converged RBM checkpoint and fit a ViT to represent the same quantum state by minimizing infidelity [23, 7]. The two represented states have fidelity $F \geq 0.98$. Thus the wavefunction is nearly fixed, while the tangent space changes. Under this matched-state protocol, ViT's larger tangent space lowers the expressivity gap variance from $\sigma^2_{\text{gap}} = 1.95$ to 0.816. The excess-risk estimate also decreases from $\mathcal{E}_\lambda \approx 2.53 \pm 0.18$ to $\mathcal{E}_\lambda \approx 1.59 \pm 0.079$. Thus overparameterization can improve SR generalization when its additional tangent directions reduce enough misspecification noise.

Together, these comparisons show the central tension of overparameterized SR. A larger model improves the update by reducing the expressivity gap, but it also worsens the update when a comparable gap remains and the finite-sample solve overfits it. This emphasizes the role of the diagonal shift λ as a statistical filter that controls how much sampled gap noise enters the SR update.

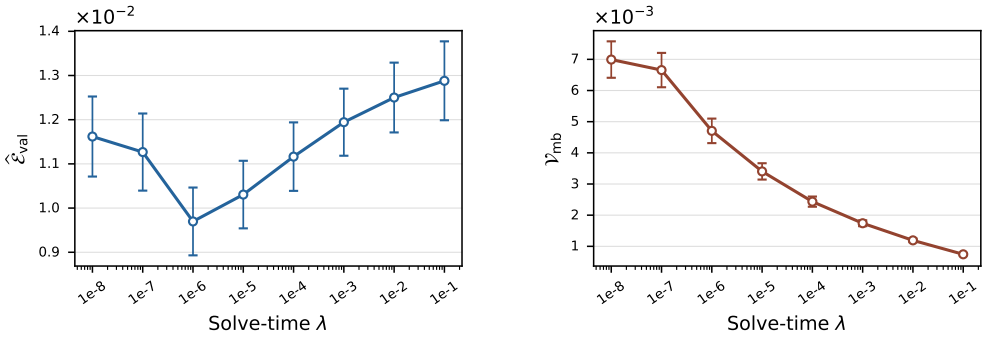


Figure 3: Large-scale shared-checkpoint test of the noisy-ridge mechanism on the periodic $L = 100$ transverse-field Ising model with h ranging from 0.8 to 1.2 with foundational NQS. A fixed set of late-training checkpoints is reused while only the solve-time shift λ is varied. $\widehat{\mathcal{E}}_{\mathrm{val}}$ is U-shaped, while $\mathcal{V}_{\mathrm{mb}}$ decreases monotonically with λ, matching the two predictions of Section 2. Error bars are one standard error across the ten checkpoint-level means.

4 Large-Scale Checkpoint Diagnostics of the Noisy-Ridge Mechanism

The exact 4×4 study isolates the noisy-ridge mechanism using population quantities. We next test whether the same mechanism remains visible in modern overparameterized NQS, where the population QGT, population SR direction, and expressivity-gap variance are unavailable. We therefore test the mechanism using the checkpoint-local diagnostics introduced in Section 2.

We use a foundation neural quantum state (FNQS) on the $L = 100$ periodic one-dimensional transverse-field Ising model (TFIM) family, $H = -h \sum_i X_i - \sum_i Z_i Z_{i+1}$, with $h \in [0.8, 1.2]$. A single conditional wave function $\log \psi_\theta(x; h)$ is trained across many Hamiltonian parameters, so the SR update must aggregate information over a family of Hamiltonians. The model has $P = 198{,}144$ parameters, while each SR step uses a batch of $N_s = 12{,}000$ samples, split over 6000 training h values with two spin configurations per field. We use the same architectural and sampler setting as Rende et al. [18] (see Appendix D for experimental details).

The diagnostic experiment uses a shared-checkpoint protocol. We first take ten late-training checkpoints from the run trained with $\lambda_{\mathrm{train}} = 10^{-4}$. These checkpoints are then held fixed. At each checkpoint, we perform an offline sweep over the solve-time shift $\lambda \in \{10^{-8}, 10^{-7}, \ldots, 10^{-1}\}$, recomputing standard SR updates on fresh resampled batches for each value of λ. This design keeps the wave function, tangent geometry, and expressivity-gap fixed while varying only the ridge penalty. It therefore tests the local bias–variance mechanism directly, not the different training trajectories.

For each checkpoint and each diagnostic shift, we construct $m = 20$ independent standard-SR updates from independently resampled training batches. The validation diagnostic uses $120{,}000$ held-out samples, consisting of 6000 held-out h values, disjoint from the training h values, and 20 spin configurations per h value. Predictions are centered separately for each h value, since each Hamiltonian parameter has its own local-energy distribution. The validation pass then estimates both $\widehat{\mathcal{E}}_{\mathrm{val}}$ and $\mathcal{V}_{\mathrm{mb}}$ without forming or solving any validation QGT system.

Figure 3 shows the fixed-checkpoint signature predicted by the noisy-ridge view. The $\hat{\mathcal{E}}_{\mathrm{val}}$ curve is U-shaped in λ, while $\mathcal{V}_{\mathrm{mb}}$ decreases as λ increases. Since all shifts are evaluated on the same checkpoints, this trend reflects the solve-time ridge filter rather than different training trajectories. Small shifts leave useful directions nearly undamped but fit more batch-dependent noise. Large shifts reduce this variance but also shrink useful population directions.

5 Multi-Shift SR for Statistical Filtering and Variance Reduction

Motivated by the noisy-ridge view, we introduce multi-shift SR (MS-SR), an optimizer that replaces the single shifted SR solve with a mixture of shifted SRs computed on independent batches. Standard SR uses one ridge scale and therefore applies the same spectral filter $s/(s+\lambda)$ to every QGT eigendi-

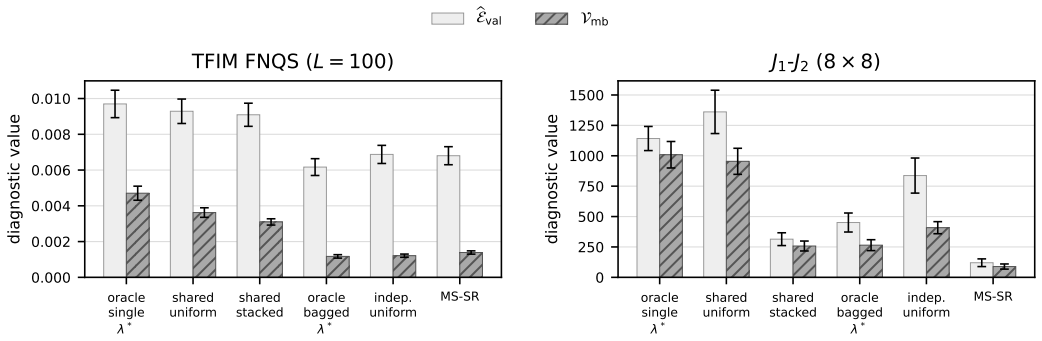


Figure 4: Checkpoint-local ablations for the ingredients of MS-SR at $K = 4$. Paired bars show raw $\widehat{\mathcal{E}}_{\mathrm{val}}$ and $\mathcal{V}_{\mathrm{mb}}$ averaged over ten fixed source checkpoints; error bars are one standard error across checkpoints. Both single SR and bagged SR are *evaluation-oracle* baselines using the validation-optimal λ^* at each checkpoint. MS-SR selects its NTK-quantile shift grid without validation tuning.

rection. MS-SR instead combines several regularization scales, while the independent batches decorrelate finite-sample errors.

At checkpoint θ_t, MS-SR draws K independent batches $\mathcal{D}_1, \ldots, \mathcal{D}_K$, chooses shifts λ_k from empirical NTK-spectrum quantiles, and computes candidate updates $\hat{\delta}_k = \mathrm{minSR}(\mathcal{D}_k, \lambda_k)$. It then forms $\hat{\delta}_{\mathrm{MS}} = \sum_{k=1}^{K} w_k \hat{\delta}_k$, with $\mathbf{w} \in \Delta^{K-1}$. The weights are selected by leave-one-batch-out stacking, so each candidate update is evaluated only on batches that were not used to compute it. The full procedure is given in Algorithm 1 in Appendix E.

This construction has a simple bias–variance interpretation. For fixed shifts and mixture weights, define $\bar{\delta}_k = \mathbb{E}_{\mathcal{D}_k}\left[\hat{\delta}_k\right]$. Then the excess risk decomposes exactly as

$$\mathbb{E}\left[\left\|\hat{\delta}_{\mathrm{MS}} - \delta^*\right\|_S^2\right] = \left\|\sum_{k=1}^{K} w_k \bar{\delta}_k - \delta^*\right\|_S^2 + \mathbb{E}\left[\left\|\sum_{k=1}^{K} w_k \left(\hat{\delta}_k - \bar{\delta}_k\right)\right\|_S^2\right]. \tag{6}$$

The first term is the bias of the average filter. The second term is the finite-batch variance of the average update. MS-SR targets both terms: multiple shifts enlarge the filter class, while independent batches reduce the variance of the noisy solves.

Why multiple shifts help. Consider the population spectral approximation from Section 2. If $S = V\,\mathrm{diag}(s_i)V^T$, then a single ridge solve applies the filter $f_\lambda(s) = s/(s+\lambda)$ to each eigendirection. A convex mixture of ridge solves applies instead $f_{\mathbf{w},\boldsymbol{\lambda}}(s) = \sum_k w_k s/(s+\lambda_k)$. Thus MS-SR is still a spectral filter, but it is no longer restricted to the single-shift family.

The difference is clearest at the two ends of the spectrum. For small s, $f_{\mathbf{w},\boldsymbol{\lambda}}(s) = s\sum_k w_k/\lambda_k + O(s^2)$. For large s, $f_{\mathbf{w},\boldsymbol{\lambda}}(s) = 1 - (\sum_k w_k\lambda_k)/s + O(s^{-2})$. With a single shift, the same λ controls both the suppression of low-eigenvalue directions and the residual shrinkage of high-eigenvalue directions. On the other hand, these two behaviors can be tuned separately with multiple shifts. This can improve the aggregate bias–variance tradeoff when useful update directions and noise-sensitive directions are not controlled well by the same shift.

Why independent batches help. The second term in Eq. (6) depends on the correlations between candidate updates. If the candidates are computed on independent batches, the cross terms vanish and $\mathbb{E}\|\sum_k w_k(\hat{\delta}^{(k)}_{\lambda_k} - \bar{\delta}_{\lambda_k})\|_S^2 = \sum_k w_k^2 \mathbb{E}\|\hat{\delta}^{(k)}_{\lambda_k} - \bar{\delta}_{\lambda_k}\|_S^2$. For uniform weights, this is a factor-K reduction of the average single-batch variance in the ideal independent-batch limit.

Independent batches produce a bagging-like effect by decorrelating the sampling noise across candidate updates. If all K shifts are solved on the same batch, the resulting candidates can still enrich the spectral filter, but their sampling errors remain shared. Consequently, the variance-reduction effect is lost. A single larger batch of size KN_s would also reduce variance, but it is substantially more

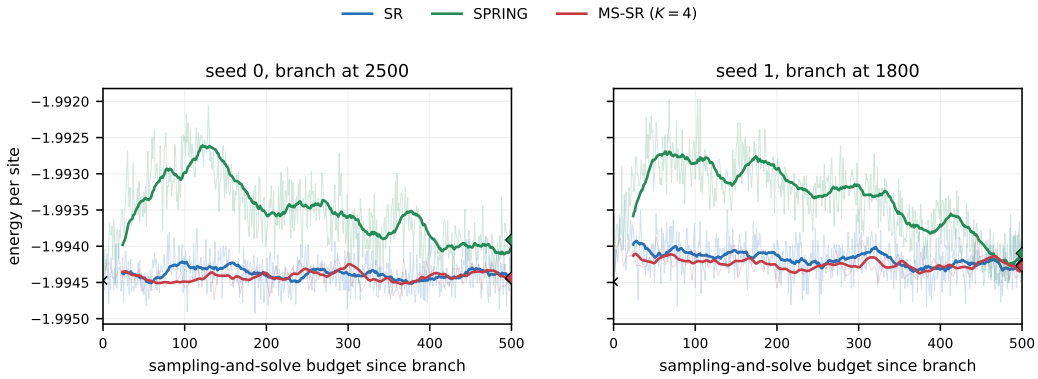


Figure 5: Equal-budget late-stage online comparison on the 8×8 J_1-J_2 model with a no-symmetry ViT. Each panel branches SR, SPRING, and fixed-grid MS-SR from identical parameters. One SR or SPRING update is one budget unit; one $K = 4$ MS-SR update is four. Faint lines are raw training-energy estimates and solid lines are rolling means over a common 25-budget-unit window. Diamonds are audited fresh-chain endpoint evaluations; bars are one conservative reported Monte Carlo standard error, not variation across seeds.

expensive computationally, with cost $O(K^3 N_s^3)$. On the other hand, K independent solves at batch size N_s cost only $O(K N_s^3)$ and can be run in parallel. MS-SR obtains the statistical benefit of a larger batch without paying the cubic cost of a single enlarged solve.

Shift and weight selection. MS-SR first chooses the shifts and then fits the mixture weights. In our checkpoint-local $K = 4$ experiments, the shifts are the empirical NTK-spectrum quantiles $\{0.9, 0.7, 0.4, 0.1\}$. These quantiles place the ridge transition scale at several parts of the empirical spectrum, producing candidate updates with different bias–variance tradeoffs rather than relying on a single regularization. The online comparison instead uses the predeclared fixed grid $\{10^{-2}, 10^{-3}, 10^{-4}, 10^{-5}\}$.

Given these shifts, Algorithm 1 uses leave-one-batch-out stacking to choose the weights. Each candidate update is evaluated on batches that were not used to compute it. The weights minimize the held-out residual between tangent-feature predictions and centered local energies, giving a direct estimate of the regression risk from Eq. (4).

Ablation study of MS-SR. Figure 4 compares six checkpoint-local updates using ten fixed checkpoints from TFIM and from the 8×8 square-lattice J_1-J_2 model at $J_2/J_1 = 0.5$. The single and bagged baselines both use the per-checkpoint validation minimizer λ^* over $\{10^{-8}, \ldots, 10^{-1}\}$; bagging averages four independent solves at that same λ^*. These are deliberately strong diagnostic oracles, not online methods. The shared-batch variants isolate multi-shift filtering, the independent-batch variants isolate averaging, and MS-SR adds stacking.

The ordering is regime-dependent. On TFIM, where $\lambda^* = 10^{-6}$ at all ten checkpoints, oracle bagging has lower validation risk than MS-SR (0.00617 versus 0.00680). At the training shift 10^{-4}, however, bagging has risk 0.00933, and MS-SR is lower on all ten checkpoints. On the harder J_1-J_2 checkpoints, MS-SR has risk 120.1, versus 451.0 for oracle bagging, a 73% reduction. Thus the additional shifts are most useful when no stable single shift suffices; when the validation-optimal shift is stable, single-shift bagging can be preferable. Appendix E.1 reports the fixed-shift, K, quantile, and source-shift studies, and Appendix E.3 reports measured wall time.

Equal-budget online comparison. Figure 5 compares two controlled late-stage branches at equal sampling-and-solve budget. Each arm receives 500 units, so SR and SPRING take 500 updates while MS-SR takes 125. For seed 0 (branch step 2500), MS-SR $-1.9944365 \pm 7.06 \times 10^{-6}$, SR $-1.9944498 \pm 6.86 \times 10^{-6}$, SPRING $-1.9939130 \pm 8.36 \times 10^{-6}$. The numerical endpoint ordering (lower is better) is SR $<$ MS-SR $<$ SPRING; MS-SR minus SR is $+0.0000133$ per site ($+1.36$ combined endpoint Monte Carlo standard errors); MS-SR minus SPRING is -0.0005235 per site (-47.84 combined endpoint Monte Carlo standard errors). For seed 1 (branch step 1800), MS-SR

$-1.9942723 \pm 7.83 \times 10^{-6}$, SR $-1.9942534 \pm 7.88 \times 10^{-6}$, SPRING $-1.9940982 \pm 8.02 \times 10^{-6}$. The numerical endpoint ordering (lower is better) is MS-SR $<$ SR $<$ SPRING; MS-SR minus SR is -0.0000189 per site (-1.70 combined endpoint Monte Carlo standard errors); MS-SR minus SPRING is -0.0001741 per site (-15.53 combined endpoint Monte Carlo standard errors). These are two individual branched trajectories; the endpoint uncertainties measure Monte Carlo evaluation error and do not estimate across-seed variation.

6 Related Work

Neural quantum states have evolved rapidly from compact RBM-like ansätze to deep convolutional, recurrent, transformer-based, fermionic, and foundation-style wave functions [2, 10, 17, 30, 15, 19, 16, 18]. Autoregressive NQS provide a complementary route to scalability because their normalized factorization permits sampling from the model Born distribution rather than Markov-chain sampling [10]. This changes the sampling and autocorrelation structure of VMC, but it does not remove the finite-sample nature of SR, since QGTs and forces are still estimated from finite samples.

SR is the natural-gradient method of VMC, with the QGT playing the role of the Fisher metric on the variational manifold [24, 25, 2]. Recent sample-space formulations, including minSR, avoid forming the full $P \times P$ QGT and instead solve an $N_s \times N_s$ kernel system, making SR practical for highly overparameterized NQS [3, 19]. SPRING builds on this view with a Kaczmarz-inspired projected-increment update that reuses information across minibatches to accelerate convergence [5, 6]. These methods improve the cost and convergence of SR, while our focus is the statistical filtering role of the diagonal shift.

A parallel line of work approximates the QGT or Fisher metric structurally. KFAC uses Kronecker-factored curvature approximations and has been important in FermiNet-style electronic-structure VMC [13, 17]. Block-diagonal QGT methods partition the NQS metric by network layers, preserving intra-layer curvature while suppressing noisy cross-layer correlations [22]. These approaches change the metric approximation. MS-SR is complementary: it keeps the sample-space SR geometry but replaces the single-shift ridge filter with a data-adaptive mixture of shifted solves.

Our analysis connects SR to random-design ridge regression [11]. In this view, the local-energy target is generally misspecified by the current tangent space, and the expressivity gap acts as the irreducible residual noise. Sketch-and-project and Kaczmarz interpretations clarify the structure of the overparameterized linear solve [27, 5, 6]. The present work identifies the statistical source of the noise being fitted and uses that view to design richer spectral filters for SR.

7 Conclusion

Modern NQS optimization has entered a finite-sample, parameter-rich regime. In this regime the diagonal shift in SR is not just a safeguard against an ill-conditioned empirical QGT. It is also the spectral regularizer that determines how much of the sampled tangent-space target is trusted. By recasting a single SR step as misspecified ridge regression, we identified the tangent-space expressivity gap as the effective noise source: it is orthogonal to the tangent features in population, but it is present in every finite batch and can be fitted by overparameterized empirical solves.

This viewpoint explains the apparent tension observed throughout the paper. Larger models improve SR when their tangent spaces reduce the expressivity gap, but they can also amplify finite-sample overfitting when a comparable gap remains. The exact small-system experiments and the large fixed-checkpoint diagnostics show the same bias–variance structure across scales.

The filter perspective also leads to MS-SR, which mixes independent ridge solves at multiple shifts. Independent batches reduce variance, multiple shifts enlarge the filter class, and stacking adapts the mixture to the local checkpoint. The equal-budget online branches test whether this local advantage translates to training outcomes late in optimization. Across the two tested branches, MS-SR closely tracks SR, with opposite numerical ordering, and ends below SPRING in both; these are trajectory-level outcomes rather than a population optimizer ranking. The checkpoint ablations identify counter-regimes in which an oracle single shift is better. For the eigensolver implementation studied here, $K = 4$ bagged SR and MS-SR take $4.00\times$ and $4.04\times$ the wall time of single

SR, respectively; the method-specific overhead beyond the common four batches and solves is about 1%.

Limitations. The theory is checkpoint-local and its factor-K variance statement requires independent candidate batches; autocorrelation reduces the effective sample size, and shared chains can correlate candidates. The online study covers two late branches of one 8×8 system rather than a broad optimizer benchmark, and MS-SR is not uniformly superior: oracle bagging wins on TFIM, and the Shastry–Sutherland negative result in Appendix E.4 shows failure when the shift grid is misspecified. MS-SR also spends K times the sampling and solve budget of single SR. The timing ratios come from one H100 hardware/software stack and need not transfer unchanged. The evaluation-oracle baselines intentionally select λ^* on the same validation stream used to report risk, so they contain selection leakage. Endpoint error bars are finite-round MCMC standard errors and do not represent between-branch variation. These results therefore support a diagnostic, regime-dependent filtering rule rather than a universal optimizer ranking.

Overall, overparameterization should not be avoided in NQS, but it should be treated with the right statistical regularization. Treating SR as a statistical filtering problem provides both a diagnostic language for understanding existing optimizers and a design principle for building more reliable ones.

References

[1] Manuela Capello, Federico Becca, Michele Fabrizio, Sandro Sorella, and Erio Tosatti. Variational description of Mott insulators. *Physical Review Letters*, 94:026406, 2005. doi: 10.1103/PhysRevLett.94.026406.

[2] Giuseppe Carleo and Matthias Troyer. Solving the quantum many-body problem with artificial neural networks. *Science*, 355(6325):602–606, 2017.

[3] Ao Chen and Markus Heyl. Empowering deep neural quantum states through efficient optimization. *Nature Physics*, 20:1476–1481, 2024.

[4] Kenny Choo, Antonio Mezzacapo, and Giuseppe Carleo. Fermionic neural-network states for ab-initio electronic structure. *Nature Communications*, 11:2368, 2020.

[5] Gil Goldshlager, Nilin Abrahamsen, and Lin Lin. A kaczmarz-inspired approach to accelerate the optimization of neural network wavefunctions. *Journal of Computational Physics*, 511: 113351, 2024.

[6] Gil Goldshlager, Jiang Hu, and Lin Lin. A sketch-and-project analysis of subsampled natural gradient algorithms. *arXiv preprint arXiv:2508.21022*, 2025.

[7] Luca Gravina, Vincenzo Savona, and Filippo Vicentini. Neural projected quantum dynamics: a systematic study. *Quantum*, 9:1803, 2025. doi: 10.22331/q-2025-07-22-1803.

[8] Iván L. Gutiérrez and Christian B. Mendl. Real time evolution with neural-network quantum states. *Quantum*, 6:627, 2022. doi: 10.22331/q-2022-01-20-627.

[9] Jan Hermann, Zeno Schätzle, and Frank Noé. Deep-neural-network solution of the electronic Schrödinger equation. *Nature Chemistry*, 12:891–897, 2020.

[10] Mohamed Hibat-Allah, Martin Ganahl, Lauren E. Hayward, Roger G. Melko, and Juan Carrasquilla. Recurrent neural network wave functions. *Physical Review Research*, 2:023358, 2020.

[11] Daniel Hsu, Sham M Kakade, and Tong Zhang. Random design analysis of ridge regression. *Foundations of Computational Mathematics*, 14:569–600, 2014.

[12] Jane Kim, Gabriel Pescia, Bryce Fore, Jannes Nys, Giuseppe Carleo, Stefano Gandolfi, Morten Hjorth-Jensen, Alessandro Lovato, et al. Neural-network quantum states for ultra-cold Fermi gases. *Communications Physics*, 7:148, 2024.

[13] James Martens and Roger Grosse. Optimizing neural networks with kronecker-factored approximate curvature. In *Proceedings of the 32nd International Conference on Machine Learning*, volume 37 of *Proceedings of Machine Learning Research*, pages 2408–2417. PMLR, 2015. URL `https://proceedings.mlr.press/v37/martens15.html`.

[14] W. L. McMillan. Ground state of liquid He^4. *Physical Review*, 138:A442–A451, 1965. doi: 10.1103/PhysRev.138.A442.

[15] Yusuke Nomura and Masatoshi Imada. Dirac-type nodal spin liquid revealed by refined quantum many-body solver using neural-network wave function, correlation ratio, and level spectroscopy. *Physical Review X*, 11(3):031034, 2021.

[16] Rajah P Nutakki, Shokry Ahmedov, and Filippo Vicentini. Design principles of deep translationally-symmetric neural quantum states for frustrated magnets. *Physical Review Research*, 7:043099, 2025.

[17] David Pfau, James S. Spencer, Alexander G. D. G. Matthews, and W. M. C. Foulkes. Ab initio solution of the many-electron Schrödinger equation with deep neural networks. *Physical Review Research*, 2:033429, 2020.

[18] Riccardo Rende, Luciano Loris Viteritti, Federico Becca, Antonello Scardicchio, Alessandro Laio, and Giuseppe Carleo. Foundation neural-networks quantum states as a unified ansatz for multiple hamiltonians. *Nature Communications*, 16:7213, 2025.

[19] Riccardo Loris Rende, Lorenzo Luciano Viteritti, Lorenzo Bardone, Federico Becca, and Sebastian Goldt. A simple linear algebra identity to optimize large-scale neural network quantum states. *Communications Physics*, 7:260, 2024.

[20] Hiroki Saito. Solving the Bose-Hubbard model with machine learning. *Journal of the Physical Society of Japan*, 86:093001, 2017. doi: 10.7566/JPSJ.86.093001.

[21] Hiroki Saito and Masaya Kato. Machine learning technique to find quantum many-body ground states of bosons on a lattice. *Journal of the Physical Society of Japan*, 87:014001, 2018. doi: 10.7566/JPSJ.87.014001.

[22] Ahmedeo Shokry, Alessandro Santini, and Filippo Vicentini. When less is more: Approximating the quantum geometric tensor with block structures. *arXiv preprint arXiv:2510.08430*, 2025.

[23] Alessandro Sinibaldi, Clemens Giuliani, Giuseppe Carleo, and Filippo Vicentini. Unbiasing time-dependent variational monte carlo by projected quantum evolution. *Quantum*, 7:1131, 2023.

[24] Sandro Sorella. Green function Monte Carlo with stochastic reconfiguration. *Physical Review Letters*, 80(20):4558, 1998.

[25] Sandro Sorella. Generalized Lanczos algorithm for variational quantum Monte Carlo. *Physical Review B*, 64(2):024512, 2001.

[26] Kyle Sprague and Stefanie Czischek. Variational Monte Carlo with large patched transformers. *Communications Physics*, 7:90, 2024.

[27] Thomas Strohmer and Roman Vershynin. A randomized Kaczmarz algorithm with exponential convergence. *Journal of Fourier Analysis and Applications*, 15:262–278, 2009.

[28] Daisuke Tahara and Masatoshi Imada. Variational Monte Carlo method combined with quantum-number projection and multi-variable optimization. *Journal of the Physical Society of Japan*, 77:114701, 2008. doi: 10.1143/JPSJ.77.114701.

[29] Filippo Vicentini, Damian Hofmann, Attila Szabó, Dian Wu, Christopher Roth, Clemens Giuliani, Gabriel Pescia, Jannes Nys, Vladimir Vargas-Calderón, Nikita Astrakhantsev, and Giuseppe Carleo. NetKet 3: Machine learning toolbox for many-body quantum systems. *SciPost Physics Codebases*, page 7, 2022. doi: 10.21468/SciPostPhysCodeb.7.

[30] Ingrid von Glehn, James S. Spencer, and David Pfau. A self-attention ansatz for ab-initio quantum chemistry. In *The Eleventh International Conference on Learning Representations*, 2023. URL `https://openreview.net/forum?id=xveTeHVlF7j`.

[31] Hisatoshi Yokoyama and Hiroyuki Shiba. Variational Monte-Carlo studies of Hubbard model. I. *Journal of the Physical Society of Japan*, 56(4):1490–1506, 1987. doi: 10.1143/JPSJ.56.1490.

[32] Remmy Zen, Long My, Ryan Tan, Frédéric Hébert, Mario Gattobigio, Christian Miniatura, Dario Poletti, and Stéphane Bressan. Finding quantum critical points with neural-network quantum states. In *ECAI 2020: 24th European Conference on Artificial Intelligence*, volume 325, pages 1962–1969, 2020. doi: 10.3233/FAIA200315.

A Code and Data Availability

The public repository containing the code, compact artifacts, and figure regeneration tools for this work is `https://github.com/takh04/sr_filter`. The experimental tools were developed on top of NetKet, an open-source neural-quantum-state and variational Monte Carlo library distributed under the Apache License 2.0 [29].

B Fixed-Checkpoint SR Regression Identities

This appendix gives the derivations behind Section 2. All statements are checkpoint-local: the wavefunction ψ_θ, sampling distribution π, QGT, and tangent features are held fixed while only the Monte Carlo batch used to estimate the SR direction varies. We write $\|u\|_S^2 = u^T S u$. Since S may be singular, null QGT directions do not change tangent predictions and have zero S-seminorm. Throughout, † denotes the Moore–Penrose pseudoinverse; Hermitian adjoints are written explicitly as $\overline{u}^T$.

B.1 Real and complex conventions

The main text assumes real wavefunctions and real tangent features in the sampling basis. Then

$$S = \mathbb{E}_\pi[O_c(x) O_c(x)^T], \qquad g = \mathbb{E}_\pi[O_c(x) H_{\mathrm{loc},c}(x)].$$

The fixed-checkpoint population problem is the ordinary least-squares problem

$$\min_\delta \mathbb{E}_\pi\left[\left(O_c(x)^T\delta - H_{\mathrm{loc},c}(x)\right)^2\right],$$

and its minimum-norm solution is $\delta^* = S^\dagger g$.

For complex wavefunctions, let

$$O_c(x) = \nabla_\theta \log\psi_\theta(x) - \mathbb{E}_\pi[\nabla_\theta \log\psi_\theta], \qquad H_{\mathrm{loc},c}(x) = H_{\mathrm{loc}}(x) - \mathbb{E}_\pi H_{\mathrm{loc}}.$$

The Hermitian QGT and force are

$$Q = \mathbb{E}_\pi[\overline{O_c(x)}\, O_c(x)^T], \qquad f = \mathbb{E}_\pi[\overline{O_c(x)}\, H_{\mathrm{loc},c}(x)].$$

For complex parameter increments $\delta \in \mathbb{C}^p$, the regression loss is $\mathbb{E}_\pi|O_c(x)^T\delta - H_{\mathrm{loc},c}(x)|^2$, the normal equations are $Q\delta = f$, and the corresponding seminorm is $\overline{\delta}^T Q\delta = \mathbb{E}_\pi|O_c(x)^T\delta|^2$. For real parameter increments in a complex wavefunction, minimizing the same modulus-squared loss over $\delta \in \mathbb{R}^p$ gives $\mathrm{Re}[Q\,\delta] = \mathrm{Re}[f]$. Equivalently, one stacks the real and imaginary residuals and solves an ordinary real least-squares problem. Thus the real-parameter SR equation uses $S = \mathrm{Re}[Q]$ and $g = \mathrm{Re}[f]$. The derivations below are written in the real convention. In the complex convention, replace scalar squares by modulus squares and replace the S-seminorm by the corresponding real or Hermitian QGT seminorm.

B.2 Validation and variance identities

Let δ^* be the population least-squares SR direction and define

$$\epsilon(x) = H_{\mathrm{loc},c}(x) - O_c(x)^T\delta^*.$$

The population normal equations give $\mathbb{E}_\pi[O_c(x)\epsilon(x)] = 0$. Hence, for any fixed direction δ,

$$\begin{aligned}\mathbb{E}_\pi\left[\left(O_c(x)^T\delta - H_{\mathrm{loc},c}(x)\right)^2\right] &= \mathbb{E}_\pi\left[\left(O_c(x)^T(\delta-\delta^*) - \epsilon(x)\right)^2\right] \\ &= \|\delta-\delta^*\|_S^2 + \sigma_{\mathrm{gap}}^2,\end{aligned}$$

where $\sigma_{\mathrm{gap}}^2 = \mathbb{E}_\pi[\epsilon(x)^2]$. Thus an independent held-out residual estimates the S-excess risk up to the constant expressivity-gap term.

For a random diagnostic update $Z = \hat{\delta}_\lambda$ computed by resampling an SR batch at the same checkpoint, let $\mu = \mathbb{E}_\mathcal{D} Z$. Adding and subtracting μ gives

$$\mathcal{E}_\lambda = \mathbb{E}_\mathcal{D}\,\|Z-\delta^*\|_S^2 = \|\mu-\delta^*\|_S^2 + \mathbb{E}_\mathcal{D}\,\|Z-\mu\|_S^2\,.$$

The cross term vanishes because $\mathbb{E}_{\mathcal{D}}[Z-\mu]=0$. This is the usual squared-bias plus variance decomposition, with distances measured in the checkpoint QGT seminorm.

Finally, let $Z_1,\dots,Z_m$ be independent copies of Z and $\bar{Z}=m^{-1}\sum_j Z_j$. The standard sample-variance identity holds in the same seminorm:

$$\mathbb{E}\left[\frac{1}{m-1}\sum_{j=1}^{m}\left\|Z_j-\bar{Z}\right\|_S^2\right]=\mathbb{E}\left\|Z-\mu\right\|_S^2.$$

Equivalently,

$$\frac{1}{m-1}\sum_{j=1}^{m}\left\|Z_j-\bar{Z}\right\|_S^2=\frac{1}{\binom{m}{2}}\sum_{a<b}\frac{1}{2}\left\|Z_a-Z_b\right\|_S^2.$$

Replacing each S-seminorm prediction norm by its empirical held-out average $|\mathcal{D}_{\mathrm{val}}|^{-1}\sum_x(O_c(x)^T\cdot)^2$ gives the multi-batch diagnostic used in the main text.

B.3 Spectral filter proxy

On the non-null QGT subspace, write $S=V\,\mathrm{diag}(s_i)V^T$ with $s_i>0$, and let $\beta_i^*=(V^T\delta^*)_i$. The population ridge direction is

$$\delta_\lambda=(S+\lambda I)^{-1}g=(S+\lambda I)^{-1}S\delta^*.$$

Therefore

$$\delta_\lambda-\delta^*=-\lambda(S+\lambda I)^{-1}\delta^*,$$

and the exact population shrinkage bias is

$$\|\delta_\lambda-\delta^*\|_S^2=\sum_i s_i\left(\frac{\lambda}{s_i+\lambda}\right)^2(\beta_i^*)^2.$$

The scalar variance term in (2) is a reference calculation, not an assumption needed by the diagnostics. To isolate the residual-noise contribution, write

$$\hat{g}=\hat{S}\delta^*+\hat{\xi},\qquad \hat{\xi}=\frac{1}{N_s}\sum_{j=1}^{N_s}O_c(x_j)\epsilon(x_j),\qquad \mathbb{E}[\hat{\xi}]=0.$$

If one ignores the covariance-estimation fluctuation in $\hat{S}$ and replaces the empirical inverse by $A_\lambda=(S+\lambda I)^{-1}$, then

$$\hat{\delta}_\lambda-\delta_\lambda\approx A_\lambda\hat{\xi}.$$

Let $\Sigma_\epsilon=\mathbb{E}_\pi[O_c(x)O_c(x)^T\epsilon(x)^2]$. The force-noise proxy is then

$$\mathbb{E}_{\mathcal{D}}\left\|\hat{\delta}_\lambda-\delta_\lambda\right\|_S^2\approx\frac{1}{N_s}\,\mathrm{tr}[SA_\lambda\Sigma_\epsilon A_\lambda].$$

Under the scalar, homoscedastic approximation $\Sigma_\epsilon\approx\sigma_{\mathrm{gap}}^2 S$, this becomes

$$\mathbb{E}_{\mathcal{D}}\left\|\hat{\delta}_\lambda-\delta_\lambda\right\|_S^2\approx\frac{\sigma_{\mathrm{gap}}^2}{N_s}\,\mathrm{tr}[A_\lambda SA_\lambda S]=\frac{\sigma_{\mathrm{gap}}^2}{N_s}\sum_i\left(\frac{s_i}{s_i+\lambda}\right)^2.$$

B.4 Local Fubini–Study metric

Let $|\Psi(\theta)\rangle$ denote the normalized variational state and let $P_\perp=I-|\Psi(\theta)\rangle\langle\Psi(\theta)|$. The real QGT is the pullback of the Fubini–Study metric,

$$S_{ij}=\mathrm{Re}\,\langle P_\perp\partial_i\Psi(\theta),P_\perp\partial_j\Psi(\theta)\rangle.$$

In the real sampling-basis convention, this equals $\mathbb{E}_\pi[O_{c,i}(x)O_{c,j}(x)]$. A Taylor expansion of the normalized state therefore gives, for nearby displacements a and b,

$$1-|\langle\Psi(\theta+a)\mid\Psi(\theta+b)\rangle|^2=(a-b)^TS(a-b)+O\big((\|a\|+\|b\|)^3\big).$$

Taking $a=-\eta\delta$ and $b=-\eta\delta'$ gives

$$\mathcal{I}_\theta(\delta,\delta')=\eta^2(\delta-\delta')^TS(\delta-\delta')+O(\eta^3)$$

for bounded update directions. Hence the S-norm excess risk controls the expected local infidelity between the population SR update and the empirical SR update, up to the factor η^2 and higher-order terms.

B.5 Sample centering

Practical SR uses sample-centered log-derivatives and sample-centered local energies. This is exactly the ridge-regression problem with an unpenalized intercept,

$$\min_{a,\delta} \frac{1}{n} \sum_{j=1}^{n} \left(a + O(x_j)^T \delta - H_{\mathrm{loc}}(x_j)\right)^2 + \lambda \|\delta\|_2^2.$$

For fixed δ, the optimal intercept is $a = \bar{H}_{\mathrm{loc}} - \bar{O}^T \delta$. Substitution gives

$$\frac{1}{n} \sum_{j=1}^{n} \left((O(x_j) - \bar{O})^T \delta - (H_{\mathrm{loc}}(x_j) - \bar{H}_{\mathrm{loc}})\right)^2 + \lambda \|\delta\|_2^2.$$

Thus sample centering is simply the intercept form of the same regression. In population, the intercept removes only the constant energy/normalization component. It does not remove the expressivity gap, because

$$\epsilon(x) = H_{\mathrm{loc},c}(x) - O_c(x)^T \delta^*$$

has $\mathbb{E}_\pi[\epsilon(x)] = 0$ and $\mathbb{E}_\pi[O_c(x)\epsilon(x)] = 0$. A finite-sample intercept also removes the sample mean of the residual, replacing ϵ_j by $\epsilon_j - \bar{\epsilon}$, but it cannot fit any x-dependent part of the gap. These centering terms are lower-order finite-sample fluctuations and do not change the checkpoint-local bias–variance identities.

C Experimental Details: Small-Scale Experiments

System. The exact experiments in Figure 2 use the 4×4 periodic J_1-J_2 Heisenberg model with $J_1 = 1.0$ and $J_2 = 0.5$, giving full Hilbert-space dimension $2^{16} = 65{,}536$. Since all basis states are enumerable, we compute the Born distribution $\pi(x) = |\psi_\theta(x)|^2 / \|\psi_\theta\|^2$, the population QGT S, the ideal update $\delta^* = S^\dagger g$, and the expressivity-gap variance σ^2_{gap} by exact summation.

Models. The underparameterized model is a complex-valued RBM with hidden-to-visible ratio $\alpha = 2$, giving $P = 1120$ real parameters and $P/N_s = 0.273$ at the diagnostic sample size $N_s = 4096$. The overparameterized model is a complex-valued ViT with depth 2, embedding dimension 24, 4 attention heads, MLP expansion factor 4, and 2×2 patches, giving $P = 13{,}448$ real parameters and $P/N_s = 3.283$. Parameter counts are real counts, obtained by unrolling each complex parameter into real and imaginary parts.

Training and checkpoint diagnostics. The model checkpoints used in the small-scale comparisons are produced by VMC with SR, 4096 Metropolis-local samples per optimization step, learning rate 10^{-3}, and training diagonal shift 10^{-3}. Exact diagnostic summaries are saved every 10 optimizer steps. For each diagnostic checkpoint we store σ^2_{gap}, the target variance, the exact population update, and the full-Hilbert-space arrays needed to evaluate one-step SR generalization.

Matched-noise protocol. For the matched-noise comparison, the RBM and ViT are trained independently and checkpoint pairs are selected by matching the raw expressivity-gap variance σ^2_{gap}. The pair used in Figure 2 is RBM step 490 and ViT step 10, with $\sigma^2_{\mathrm{gap}} = 0.8088$ and 0.7957 respectively. At these fixed checkpoints we draw 100 independent training batches of size $N_s = 4096$ from the exact Born distribution and solve SR at $\lambda = 10^{-9}$.

Matched-state protocol. For the matched-state comparison, we first choose a partially converged RBM checkpoint and then fit a ViT to the same wavefunction by minimizing the exact Fubini–Study infidelity

$$\mathcal{I}(\theta_{\mathrm{ViT}}) = 1 - \frac{\left|\sum_x \psi^*_{\mathrm{RBM}}(x) \psi_{\mathrm{ViT}}(x; \theta_{\mathrm{ViT}})\right|^2}{\left(\sum_x |\psi_{\mathrm{RBM}}(x)|^2\right) \left(\sum_x |\psi_{\mathrm{ViT}}(x; \theta_{\mathrm{ViT}})|^2\right)}. \tag{7}$$

All sums run over the 2^{16} basis states, so the objective and diagnostics do not use stochastic overlap estimates. The pair used in Figure 2 has fidelity $F = 0.983626$ between the two represented states. Its exact gap variances are $\sigma^2_{\mathrm{gap}} = 1.9469$ for the RBM and 0.8161 for the ViT. We use 10 independent Born-resampled training batches of size 4096 for this matched-state one-step SR diagnostic.

One-step SR metrics. For each resampled training batch $\mathcal{D}_b$, the main-text excess-risk estimate uses the exact population geometry:

$$\widehat{\mathcal{E}}_\lambda = \frac{1}{B}\sum_{b=1}^{B}\left\|\hat{\delta}_\lambda(\mathcal{D}_b) - \delta^*\right\|_S^2 .$$

Here $B = 100$ for matched noise and $B = 10$ for matched state. Figure 2 and the main-text values report one sample standard error, computed as the sample standard deviation of the B trial risks divided by $\sqrt{B}$. We also compute the training-batch regression residual and, as a consistency diagnostic, the corresponding full population local-energy residual obtained by adding the constant expressivity-gap variance σ^2_{gap}.

D Experimental Details: Large-Scale Experiments

This appendix gives the reproducibility details for the two large-scale systems used in Sections 4 and 5: the $L = 100$ TFIM foundation NQS and the 8×8 J_1-J_2 ViT.

D.1 FNQS on TFIM

Hamiltonian. The periodic one-dimensional transverse-field Ising model is

$$H(h) = -h\sum_{i=1}^{L} X_i - \sum_{i=1}^{L} Z_i Z_{i+1}, \qquad L = 100,$$

with transverse field $h \in [0.8, 1.2]$. This interval straddles the critical point $h = 1$.

Foundation NQS objective. Following Rende et al. [18], we train a conditional wavefunction $\log \psi_\theta(x; h)$ rather than a separate model for each field value. The field h is supplied as a coupling token to the ViT. Training minimizes the field-averaged variational energy

$$\mathcal{L}(\theta) = \mathbb{E}_{h\sim p(h)}\left[\frac{\langle\psi_\theta(\cdot;h)|H(h)|\psi_\theta(\cdot;h)\rangle}{\langle\psi_\theta(\cdot;h)|\psi_\theta(\cdot;h)\rangle}\right], \tag{8}$$

where $p(h)$ is uniform over $R = 6000$ training fields in $[0.8, 1.2]$.

Model and optimizer. The model is a real-valued ViTFNQS with 6 layers, $d_{\mathrm{model}} = 72$, 12 attention heads, patch size $b = 4$ (effective sequence length 25), one coupling input (`n_coups=1`), and translational invariance. It has $P = 198{,}144$ trainable parameters. Each SR step uses $N_s = 12{,}000$ samples, organized as 6000 field replicas with two spin configurations per field. Spin configurations are drawn with a local Metropolis rule and sweep size L. The optimizer is SGD with fixed learning rate $\eta = 0.005$ and minSR/NTK preconditioning using the `pinv` linear solver.

Training and checkpoints. The source run used for the shared-checkpoint diagnostic protocol is trained with $\lambda_{\mathrm{train}} = 10^{-4}$ for 2000 steps. Checkpoints are saved every 100 steps. The ten diagnostic checkpoints are $\{1100, 1200, \ldots, 2000\}$.

Held-out field bank. Validation uses a fixed held-out bank of 6000 transverse fields drawn uniformly from $[0.8, 1.2]$ with seed 0, disjoint from the training fields. Each validation pass uses 20 spin configurations per held-out field, giving $|\mathcal{D}_{\mathrm{val}}| = 120{,}000$.

D.2 J_1-J_2 ViT with symmetry ramp

Hamiltonian. The two-dimensional system is the 8×8 periodic J_1-J_2 Heisenberg model

$$H = J_1\sum_{\langle i,j\rangle}\mathbf{S}_i\cdot\mathbf{S}_j + J_2\sum_{\langle\langle i,j\rangle\rangle}\mathbf{S}_i\cdot\mathbf{S}_j, \qquad J_1 = 1, \quad J_2 = 0.5.$$

No Marshall sign rule is imposed.

Model and optimizer. The ansatz is the ViT architecture of Rende et al. [19], with depth 6, $d_{\text{model}} = 48$, 8 attention heads, MLP expansion factor 4, and 2×2 patches. The real parameter count is $P = 147{,}216$. SR uses the minSR formulation with Cholesky solves, peak learning rate $\eta = 0.01$, a 100-step linear warmup, and a cosine decay schedule.

Symmetry ramp. Training uses three phases with increasing symmetry:

- identity symmetry for 200 steps with $N_s = 8192$,
- translations for 3500 steps with $N_s = 8192$,
- translations plus C_4 rotations for 3500 steps with $N_s = 6000$.

The total training length is 7200 steps. The checkpoint source used for the MS-SR ablations is the $\lambda_{\text{train}} = 10^{-4}$ run. The ten diagnostic checkpoints are steps $\{900, 1600, 2300, 3000, 3700, 4400, 5100, 5800, 6500, 7200\}$.

D.3 Checkpoint-Local Diagnostic Protocol

Solve-time shift sweep. The large-scale diagnostics hold the source checkpoint fixed and vary only the solve-time shift. For the TFIM U-curve in Figure 3, the diagnostic grid is

$$\lambda \in \{10^{-8}, 10^{-7}, 10^{-6}, 10^{-5}, 10^{-4}, 10^{-3}, 10^{-2}, 10^{-1}\}.$$

Each value recomputes SR updates from fresh training batches at the same stored parameters.

Delta banks. For each checkpoint and solve-time shift, we build a bank of independent SR updates by repeatedly restoring the checkpoint state, drawing a fresh training batch, computing the SR update, and discarding the parameter update. This makes the stored directions independent draws from the checkpoint-local law of $\hat{\delta}_\lambda$. The single-shift diagnostic banks use $m = 20$ deltas.

Validation pass. The validation diagnostics are computed by a streamed pass over a fixed validation block. Local energies and log-derivative Jacobian-vector products are computed once per validation sample and reused for all stored deltas. The J_1-J_2 validation block has 100,000 samples. The TFIM validation block has 120,000 samples, organized as 6000 held-out fields times 20 spin configurations per field.

Centering and metrics. For J_1-J_2, local energies and predictions are centered globally over the validation block. For TFIM, they are centered separately within each held-out field, matching the replica structure of the training objective. The validation residual $\widehat{\mathcal{E}}_{\text{val}}$ is the mean squared held-out regression residual for the stored updates. The multi-batch variance $\mathcal{V}_{\text{mb}}$ is the centered prediction-space disagreement of the stored updates around their bank mean; it is computed from $O_c(x)^T \hat{\delta}_j$ rather than from raw parameter differences, so flat parameter directions do not contribute.

E Offline Diagnostics and MS-SR Protocols

MS-SR shifts. All MS-SR ablations in Figure 4 use $K = 4$. The four shifts are chosen from the empirical NTK spectrum at quantiles $\{0.9, 0.7, 0.4, 0.1\}$. The quantiles are computed from one fitting batch at the current checkpoint, then reused for the candidate solves in that MS-SR step. This gives a checkpoint-adaptive grid without fitting the shifts on the final validation data.

Shared-batch and independent-batch constructions. The shared-batch variants solve all four shifts on one shared fitting batch. This preserves the exact finite-sample interpretation as a convex mixture of ridge filters on a single empirical QGT. The independent-batch variants solve one shift per independent fitting batch. They no longer correspond to a single empirical spectral filter, but their sampling errors are decorrelated, giving the variance-reduction mechanism analyzed in Section 5.

Stacking weights. For MS-SR, weights are selected by leave-one-batch-out stacking. Candidate update $\hat{\delta}_i$ is evaluated on batches that did not produce it, and the weights are constrained to the simplex. The shared-batch stacked ablation uses the same candidate deltas as the shared-batch

Algorithm 1 Multi-Shift SR (MS-SR): one optimization step

Require: parameters θ_t; number of shifts K; batch size N_s; NTK-spectrum quantiles $\{q_j\}_{j=1}^{K}$.

1: Draw independent batches $\mathcal{D}_1, \dots, \mathcal{D}_K \sim \pi_{\theta_t}$, each of size N_s. On each batch compute centered features $O_{c,j}$ and centered local energies $H_{\mathrm{loc},c,j}$.

2: Estimate one empirical NTK spectrum $\{s_i\}_{i=1}^{N_s} = \mathrm{spec}(O_{c,1}O_{c,1}^T)$, and set

$$\lambda_j \leftarrow \mathrm{Quantile}(\{s_i\}_{i=1}^{N_s}, q_j), \qquad j = 1, \dots, K.$$

3: In parallel, compute $\hat{\delta}_j \leftarrow \mathrm{minSR}(\mathcal{D}_j, \lambda_j)$ for $j = 1, \dots, K$.

4: Fit convex leave-one-batch-out stacking weights

$$\mathbf{w} \leftarrow \operatorname*{arg\,min}_{\mathbf{v} \in \Delta^{K-1}} \sum_{j=1}^{K} \left\| \sum_{i \neq j} \frac{v_i}{1 - v_j} O_{c,j} \hat{\delta}_i - H_{\mathrm{loc},c,j} \right\|^2 .$$

5: Apply the mixture update:

$$\theta_{t+1} \leftarrow \theta_t - \eta \sum_{j=1}^{K} w_j \hat{\delta}_j .$$

6: **return** $\theta_{t+1}, \boldsymbol{\lambda}, \mathbf{w}$.

uniform method, but fits the simplex weights on an independent weight-fit batch. Uniform variants set $w_i = 1/K$.

Ablations. Figure 4 compares the following six checkpoint-local updates:

- **Evaluation-oracle single SR:** one SR solve at the per-checkpoint validation minimizer λ^* over $\{10^{-8}, \dots, 10^{-1}\}$.
- **Shared-batch multi-shift, uniform:** four NTK-quantile shifts on one batch, averaged uniformly.
- **Shared-batch multi-shift, stacked:** the same four shared-batch candidates with simplex weights fit on an independent weight-fit batch.
- **Evaluation-oracle bagged SR:** four independent SR solves at the same per-checkpoint λ^*, averaged uniformly.
- **Independent-batch multi-shift, uniform:** four independent batches, one NTK-quantile shift per batch, averaged uniformly.
- **MS-SR:** four independent batches, NTK-quantile shifts, and leave-one-batch-out stacking weights.

The ablation separates the effect of richer spectral filters from the effect of independent-batch averaging. Bars in the main figure are computed from the same fixed source checkpoints for each method; error bars are standard errors over checkpoint-level diagnostic means. Figure 4 is regenerated directly from `artifacts/robustness/method_ablation_checkpoints.csv`; for each method, the script computes the checkpoint mean and the population standard deviation across the ten checkpoints divided by $\sqrt{10}$.

The earlier submission text incorrectly described both oracle baselines as using the training shift 10^{-4}. This correction makes the baselines stronger: they use data held out from update fitting and an eight-point sweep at every checkpoint, whereas MS-SR does not tune its shifts on validation. The same validation stream both selects and reports the oracle shift, so these rows deliberately contain selection leakage and are diagnostic baselines.

E.1 Baseline Tuning and Robustness

The compact inputs for this subsection are released as `artifacts/robustness/shift_sweep_summary.csv`, `artifacts/robustness/method_ablation_summary.csv`, and `artifacts/`

`robustness/bagged_fixed_shift_summary.csv`; checkpoint- and repeat-level companion files, provenance, and checksums are in the same `artifacts/robustness/` directory.

Fixed versus oracle shifts. Table 1 separates the effect of independent-batch averaging from shift selection on the ten TFIM checkpoints. The oracle rows select λ^* independently at every checkpoint from the validation residual and report risk on that same stream; this information is unavailable to an online optimizer and introduces intentional selection leakage. At the training shift, MS-SR improves validation risk on all ten paired checkpoints. At the oracle shift, bagged SR is better on TFIM. Its lower $\mathcal{V}_{\mathrm{mb}}$ at 10^{-4} does not imply lower total risk: the stronger shrinkage reduces variance while increasing bias.

Method	$\widehat{\mathcal{E}}_{\mathrm{val}}$	$\mathcal{V}_{\mathrm{mb}}$
Evaluation-oracle single SR	0.00970 ± 0.00077	0.00471 ± 0.00039
Bagged SR at $\lambda_{\mathrm{train}} = 10^{-4}$	0.00933 ± 0.00065	0.000613 ± 0.000040
Evaluation-oracle bagged SR	0.00617 ± 0.00047	0.001171 ± 0.000096
MS-SR	0.00680 ± 0.00050	0.001389 ± 0.000092

Table 1: TFIM shift comparison. Entries are means $\pm$ one standard error across ten checkpoints. The oracle rows use the same fitting-held-out validation sweep for selection and reporting at each checkpoint; MS-SR uses the fixed NTK-quantile rule.

Number of candidates and quantile grid. Table 2 varies K and the empirical-spectrum quantiles. The paper's $K = 4$ and default grid were fixed before this study. Increasing K smoothly lowers both diagnostics on TFIM. The two alternative $K = 4$ grids change validation risk by at most 3.4% relative to the default. Oracle bagging remains lower on this single-shift-sufficient system for every K tested, which is why these results should not be read as a universal MS-SR advantage.

Setting	MS-SR $\widehat{\mathcal{E}}_{\mathrm{val}}$	MS-SR $\mathcal{V}_{\mathrm{mb}}$	Oracle-bagged $\widehat{\mathcal{E}}_{\mathrm{val}}$
$K = 2$	0.008538 ± 0.000617	0.002277 ± 0.000175	0.007332 ± 0.000568
$K = 3$	0.007157 ± 0.000524	0.001738 ± 0.000109	0.006556 ± 0.000502
$K = 4$	0.006802 ± 0.000504	0.001389 ± 0.000092	0.006166 ± 0.000470
$K = 5$	0.006383 ± 0.000468	0.001266 ± 0.000087	0.005934 ± 0.000452
$K = 4$, $(.95, .8, .5, .2)$	0.006938 ± 0.000514	0.001469 ± 0.000089	—
$K = 4$, $(.8, .6, .3, .05)$	0.006574 ± 0.000496	0.001369 ± 0.000102	—

Table 2: Robustness to K and the quantile grid on ten TFIM checkpoints. Entries are means $\pm$ one standard error across checkpoints. The default $K = 4$ grid is $(.9, .7, .4, .1)$. Oracle bagging uses the validation-selected λ^* for each checkpoint.

Source training shift. We also repeat the $K = 4$ checkpoint protocol on source trajectories trained at $\lambda_{\mathrm{train}} = 10^{-3}$ and 10^{-5} (Table 3). These small three-checkpoint tests give near-parity between MS-SR and oracle bagging and therefore support robustness, not superiority.

λ_{train}	Checkpoints	MS-SR $\widehat{\mathcal{E}}_{\mathrm{val}}$	Oracle-bagged $\widehat{\mathcal{E}}_{\mathrm{val}}$
10^{-3}	3	0.02473	0.02459
10^{-5}	3	0.001575	0.001580

Table 3: Source-shift robustness. Because this follow-up contains only three checkpoints per source shift, we report descriptive means without uncertainty bars to avoid overinterpreting the small checkpoint sample.

Correlated samples and a deployment rule. Within-batch Markov-chain autocorrelation replaces N_s in the scalar proxy by an effective sample size and is already present in the empirical diagnostics. The exact factor-K reduction in Section 5, however, requires candidate errors to be independent; we therefore use separately seeded chains for the K batches. A practical rule is to

spend the K-batch budget only when $\mathcal{V}_{\mathrm{mb}}$ is a material share of the reducible validation risk and the validation-optimal shift varies across nearby checkpoints. Stable λ^* and a shallow U-curve favor single-shift bagging.

E.2 Equal-Budget Online Protocol

Figure 5 uses two independently seeded late branches of the 8×8 periodic J_1-J_2 model at $J_2/J_1 = 0.5$. Each branch starts from one common SR checkpoint and then runs SR, SPRING, and MS-SR from identical parameters. Stable, machine-readable branch metadata and the exact endpoint-evaluation counts are summarized below. The corresponding files are released under `artifacts/online_training/branch_seed0/` and `artifacts/online_training/branch_seed1/`.

Branch	Source step	Updates (SR / SPRING / MS-SR)	Endpoint samples (SR / SPRING / MS-SR)
seed 0	2500	500 / 500 / 125	8,388,608 / 8,388,608 / 8,388,608
seed 1	1800	500 / 500 / 125	8,388,608 / 8,388,608 / 8,388,608

Table 4: Online-branch accounting from the audited artifacts. Each optimizer receives 500 budget units; all candidate batches contain 8192 samples.

Each endpoint replica uses 512 evaluation rounds, 5 burn-in rounds, and 8,192 samples per round. Endpoint evaluation seeds (SR / SPRING / MS-SR) are seed 0: 12345/98765 / 12345/98765 / 12345/98765; seed 1: 12345/98765 / 12345/98765 / 12345/98765.

The no-symmetry ViT has depth 8, embedding dimension 72, 12 attention heads, MLP expansion factor 4, and 2×2 patches. Every candidate batch contains 8192 samples. Training uses a constant learning rate 0.01 and a Cholesky minSR solver. SR and SPRING use shift 10^{-4}; SPRING uses momentum 0.9. The online MS-SR variant uses $K = 4$ independent batches, exact leave-one-batch-out stacking, and the fixed shift grid $\{10^{-2}, 10^{-3}, 10^{-4}, 10^{-5}\}$. This fixed grid differs from the NTK-quantile construction used in the checkpoint-local ablations and is stated separately to avoid conflating the two protocols.

One SR or SPRING update is one budget unit, while one $K = 4$ MS-SR update is four units. Each continuation receives 500 units, hence SR and SPRING take 500 updates and MS-SR takes 125. The plotted faint curves are raw stochastic training-energy estimates; solid curves use a common 25-budget-unit rolling mean after interpolation to a one-unit budget grid. Endpoint markers use fresh, independently seeded replicas and report one standard error. Each replica resets once, advances a single Markov stream through five discarded rounds, and then retains 512 consecutive rounds of 8192 samples; it is not reseeded between rounds. Repeated endpoint evaluations are pooled only after passing the artifact audit, including a predeclared check that the replica means differ by no more than five combined standard errors. Because the two panels are individual branched trajectories rather than replicated estimates of a population effect, we do not attach across-seed error bars to their training curves.

E.3 Wall-Clock Benchmark

The direct timing benchmark uses the TFIM $L = 100$ step-2000 checkpoint with $N_s = 12{,}000$ samples per candidate batch on one NVIDIA H100 80GB HBM3 GPU, hosted by two Intel Xeon Platinum 8481C CPUs. All methods use the same Hermitian eigendecomposition pseudoinverse. Within each of three device/seed pairs, each method has two warm-ups followed by five device-synchronized repetitions. Sampler fingerprints verify that SR's batch equals the first candidate batch of both $K = 4$ methods and that all four candidate batches agree between bagged SR and MS-SR. The common parameter write and optional residual diagnostics are excluded. The long-form timings, aggregate table, and paired differences are released as `artifacts/wall_clock/timings_long.csv`, `artifacts/wall_clock/timing_summary.csv`, and `artifacts/wall_clock/paired_cluster_summary.csv`. Pairing checks are in `artifacts/wall_clock/pairing_verification.csv`; the nine sanitized per-run records, provenance, and checksums are in the same `artifacts/wall_clock/` directory.

Memory measurement. The raw records also retain JAX allocator high-water marks. They were 57.986 GB for SR and 58.124 GB for both bagged SR and MS-SR (54.004 and 54.132

GiB, respectively). These counters cover the full process and JIT-compilation lifetime rather than an isolated steady-state update, so we report them only as a feasibility check on the 80 GB device and do not interpret the 0.138 GB difference as a method-specific peak-memory estimate. A defensible comparison of incremental step memory would require a fresh process per method and allocator reset or timeline profiling; that comparison is recorded as excluded in `artifacts/wall_clock/exclusions.json`.

Method	Batches	Solves	Median time (three-pair range)	Relative to SR
SR	1	1	13.3836 s (13.1536–13.4493)	$1.0000\times$
Bagged SR ($K = 4$)	4	4	53.5488 s (52.7056–53.5987)	$3.9991\times$
MS-SR ($K = 4$)	4	4	54.0858 s (53.1442–54.1786)	$4.0389\times$

Table 5: Single-GPU steady-state wall time. We first take the median over five repetitions within each paired device/seed run, then report the median and range over three pairs; these are not standard errors.

The paired MS-SR increment over bagged SR is 0.5503 s (1.027%), with three-pair ranges 0.4373–0.5790 s and 0.829–1.080%. The dominant cost of both methods is therefore the same four batches and four solves. MS-SR reuses the first eigendecomposition for its spectrum and first shifted solve, reuses candidate batches and local energies for stacking, evaluates its twelve off-batch directions in four batched Jacobian–vector products, and adds about 1% measured method-specific overhead beyond bagging. This does not remove the approximately fourfold total cost relative to single SR.

E.4 Shastry–Sutherland Negative Result

We also ran the fixed-checkpoint protocol on a no-symmetry 8×8 Shastry–Sutherland model at $J'/J = 0.8$. This is a negative result for the shipped MS-SR rule. The boundary-grid evaluation-oracle bagged SR baseline selects $\lambda^* = 10^{-1}$—the largest tested value—on the same validation stream used to report risk, at all ten checkpoints. It obtains raw validation risk 32.8 ± 3.1, whereas MS-SR obtains 158.6 ± 38.3 (mean $\pm$ one standard error across checkpoints). Thus the boundary-grid evaluation-oracle bagged SR baseline is about five times better within this diagnostic comparison.

This system is an extreme single-shift-sufficient regime and the candidate multi-shift family is misspecified at the grid boundary. It supports the deployment rule above: when λ^* is stable and at an edge, the extra K-batch mixture is not justified. Because the optimum lies on the search boundary, the values do not establish the unconstrained best shift; they only show that the tested MS-SR construction fails against the tested oracle family. The compact checkpoint summary is released at `artifacts/robustness/ssm_boundary_oracle_summary.csv`.

F Compute Resources

Experiment	Manuscript use	Worker	Recorded accounting
Exact 4×4 studies, including discarded exploratory runs	Figure 2	$1\times$ H100 80GB GPU	$\approx$ 30 h / 30 GPU-h
TFIM-FNQS training	Figure 3 and 4 checkpoints	$8\times$ H100 80GB GPUs	$\approx$ 1.9 h / 15 GPU-h
TFIM single-shift sweep	Figure 3	$8\times$ H100 80GB GPUs	$\approx$ 25.5 h / 204 GPU-h
J_1–J_2 ViT training	Figure 4 checkpoints	$8\times$ H100 80GB GPUs	$\approx$ 15.1 h / 121 GPU-h
J_1–J_2 MS-SR ablations	Figure 4	$8\times$ H100 80GB GPUs	$\approx$ 46 h / 368 GPU-h
TFIM MS-SR ablations	Figure 4	$8\times$ H100 80GB GPUs	$\approx$ 3.0 h / 24 GPU-h
TFIM robustness follow-ups	Tables 1–3	Single H100 80GB jobs	10 checkpoints/configuration (3 for source-shift rows); end-to-end time not retained
Online branches and endpoint evaluations	Figure 5	$4\times$ H100 80GB GPUs per arm	Per branch: 500 SR, 500 SPRING, 125 MS-SR updates; endpoint sample counts in Table 4
Optimized wall-clock benchmark	Table 5	$1\times$ H100 80GB GPU	3 pairs $\times$ 3 methods $\times$ (2 warm-ups + 5 timed repeats); $<$ 1 GPU-h
Shastry–Sutherland diagnostic	Appendix E.4	Single H100 80GB jobs	10 checkpoints, 20 resampled updates/checkpoint; end-to-end time not retained

Table 6: Compute-resource accounting. The first six entries sum to approximately 762 H100 GPU-hours. This total excludes response-period follow-ups whose original launch logs did not retain complete end-to-end times; for those rows we report the exact worker and workload instead of inventing GPU-hour totals. The 4×4 row explicitly includes exploratory experiments removed from the final manuscript.

NeurIPS Paper Checklist

1. **Claims**

   Question: Do the main claims made in the abstract and introduction accurately reflect the paper's contributions and scope?

   Answer: [Yes]

   Justification: The abstract and introduction state the paper's main claims: the fixed-checkpoint ridge-regression view of SR, the role of the expressivity gap, the exact and large-scale diagnostics, the regime-dependent MS-SR ablations, and the scope of the two online branches.

   Guidelines:

   - The answer [N/A] means that the abstract and introduction do not include the claims made in the paper.
   - The abstract and/or introduction should clearly state the claims made, including the contributions made in the paper and important assumptions and limitations. A [No] or [N/A] answer to this question will not be perceived well by the reviewers.
   - The claims made should match theoretical and experimental results, and reflect how much the results can be expected to generalize to other settings.
   - It is fine to include aspirational goals as motivation as long as it is clear that these goals are not attained by the paper.

2. **Limitations**

   Question: Does the paper discuss the limitations of the work performed by the authors?

   Answer: [Yes]

   Justification: The explicit Limitations paragraph in Section 7 discusses the checkpoint-local scope, independent-batch assumption, correlated sampling, K-fold cost, limited online replication, and the reported negative result.

   Guidelines:

- The answer [N/A] means that the paper has no limitation while the answer [No] means that the paper has limitations, but those are not discussed in the paper.
- The authors are encouraged to create a separate "Limitations" section in their paper.
- The paper should point out any strong assumptions and how robust the results are to violations of these assumptions (e.g., independence assumptions, noiseless settings, model well-specification, asymptotic approximations only holding locally). The authors should reflect on how these assumptions might be violated in practice and what the implications would be.
- The authors should reflect on the scope of the claims made, e.g., if the approach was only tested on a few datasets or with a few runs. In general, empirical results often depend on implicit assumptions, which should be articulated.
- The authors should reflect on the factors that influence the performance of the approach. For example, a facial recognition algorithm may perform poorly when image resolution is low or images are taken in low lighting. Or a speech-to-text system might not be used reliably to provide closed captions for online lectures because it fails to handle technical jargon.
- The authors should discuss the computational efficiency of the proposed algorithms and how they scale with dataset size.
- If applicable, the authors should discuss possible limitations of their approach to address problems of privacy and fairness.
- While the authors might fear that complete honesty about limitations might be used by reviewers as grounds for rejection, a worse outcome might be that reviewers discover limitations that aren't acknowledged in the paper. The authors should use their best judgment and recognize that individual actions in favor of transparency play an important role in developing norms that preserve the integrity of the community. Reviewers will be specifically instructed to not penalize honesty concerning limitations.

3. **Theory assumptions and proofs**

   Question: For each theoretical result, does the paper provide the full set of assumptions and a complete (and correct) proof?

   Answer: [Yes]

   Justification: Section 2 states the fixed-checkpoint regression setup, and Appendix B gives the supporting identities and spectral-risk derivation.

   Guidelines:

   - The answer [N/A] means that the paper does not include theoretical results.
   - All the theorems, formulas, and proofs in the paper should be numbered and cross-referenced.
   - All assumptions should be clearly stated or referenced in the statement of any theorems.
   - The proofs can either appear in the main paper or the supplemental material, but if they appear in the supplemental material, the authors are encouraged to provide a short proof sketch to provide intuition.
   - Inversely, any informal proof provided in the core of the paper should be complemented by formal proofs provided in appendix or supplemental material.
   - Theorems and Lemmas that the proof relies upon should be properly referenced.

4. **Experimental result reproducibility**

   Question: Does the paper fully disclose all the information needed to reproduce the main experimental results of the paper to the extent that it affects the main claims and/or conclusions of the paper (regardless of whether the code and data are provided or not)?

   Answer: [Yes]

   Justification: Sections 3–5 and Appendices C–E.2 describe the systems, sample budgets, diagnostic construction, ablations, robustness studies, online branches, and stable artifact schemas needed to reproduce the reported comparisons.

   Guidelines:

- The answer [N/A] means that the paper does not include experiments.
- If the paper includes experiments, a [No] answer to this question will not be perceived well by the reviewers: Making the paper reproducible is important, regardless of whether the code and data are provided or not.
- If the contribution is a dataset and/or model, the authors should describe the steps taken to make their results reproducible or verifiable.
- Depending on the contribution, reproducibility can be accomplished in various ways. For example, if the contribution is a novel architecture, describing the architecture fully might suffice, or if the contribution is a specific model and empirical evaluation, it may be necessary to either make it possible for others to replicate the model with the same dataset, or provide access to the model. In general. releasing code and data is often one good way to accomplish this, but reproducibility can also be provided via detailed instructions for how to replicate the results, access to a hosted model (e.g., in the case of a large language model), releasing of a model checkpoint, or other means that are appropriate to the research performed.
- While NeurIPS does not require releasing code, the conference does require all submissions to provide some reasonable avenue for reproducibility, which may depend on the nature of the contribution. For example
  (a) If the contribution is primarily a new algorithm, the paper should make it clear how to reproduce that algorithm.
  (b) If the contribution is primarily a new model architecture, the paper should describe the architecture clearly and fully.
  (c) If the contribution is a new model (e.g., a large language model), then there should either be a way to access this model for reproducing the results or a way to reproduce the model (e.g., with an open-source dataset or instructions for how to construct the dataset).
  (d) We recognize that reproducibility may be tricky in some cases, in which case authors are welcome to describe the particular way they provide for reproducibility. In the case of closed-source models, it may be that access to the model is limited in some way (e.g., to registered users), but it should be possible for other researchers to have some path to reproducing or verifying the results.

5. **Open access to data and code**

   Question: Does the paper provide open access to the data and code, with sufficient instructions to faithfully reproduce the main experimental results, as described in supplemental material?

   Answer: [Yes]

   Justification: Appendix A links the public repository containing the code and data for this work: `https://github.com/takh04/sr_filter`. The manuscript and appendices document the experimental protocols and run settings, and the repository includes quickstart and figure-regeneration commands.

   Guidelines:

   - The answer [N/A] means that paper does not include experiments requiring code.
   - Please see the NeurIPS code and data submission guidelines (`https://neurips.cc/public/guides/CodeSubmissionPolicy`) for more details.
   - While we encourage the release of code and data, we understand that this might not be possible, so [No] is an acceptable answer. Papers cannot be rejected simply for not including code, unless this is central to the contribution (e.g., for a new open-source benchmark).
   - The instructions should contain the exact command and environment needed to run to reproduce the results. See the NeurIPS code and data submission guidelines (`https://neurips.cc/public/guides/CodeSubmissionPolicy`) for more details.
   - The authors should provide instructions on data access and preparation, including how to access the raw data, preprocessed data, intermediate data, and generated data, etc.
   - The authors should provide scripts to reproduce all experimental results for the new proposed method and baselines. If only a subset of experiments are reproducible, they should state which ones are omitted from the script and why.

- At submission time, to preserve anonymity, the authors should release anonymized versions (if applicable).
- Providing as much information as possible in supplemental material (appended to the paper) is recommended, but including URLs to data and code is permitted.

6. **Experimental setting/details**

   Question: Does the paper specify all the training and test details (e.g., data splits, hyperparameters, how they were chosen, type of optimizer) necessary to understand the results?

   Answer: [Yes]

   Justification: The main experimental sections and appendices specify the Hamiltonians, architectures, sample sizes, checkpoint selection, validation batches, shift sweeps, and MS-SR weighting rules.

   Guidelines:

   - The answer [N/A] means that the paper does not include experiments.
   - The experimental setting should be presented in the core of the paper to a level of detail that is necessary to appreciate the results and make sense of them.
   - The full details can be provided either with the code, in appendix, or as supplemental material.

7. **Experiment statistical significance**

   Question: Does the paper report error bars suitably and correctly defined or other appropriate information about the statistical significance of the experiments?

   Answer: [Yes]

   Justification: The exact experiments use full-population quantities where available, and the large-scale/MS-SR diagnostics report checkpoint-local means with explicitly identified standard errors. The online curves are identified as individual branches; their fresh endpoint evaluations use Monte Carlo standard errors rather than across-seed bars.

   Guidelines:

   - The answer [N/A] means that the paper does not include experiments.
   - The authors should answer [Yes] if the results are accompanied by error bars, confidence intervals, or statistical significance tests, at least for the experiments that support the main claims of the paper.
   - The factors of variability that the error bars are capturing should be clearly stated (for example, train/test split, initialization, random drawing of some parameter, or overall run with given experimental conditions).
   - The method for calculating the error bars should be explained (closed form formula, call to a library function, bootstrap, etc.)
   - The assumptions made should be given (e.g., Normally distributed errors).
   - It should be clear whether the error bar is the standard deviation or the standard error of the mean.
   - It is OK to report 1-sigma error bars, but one should state it. The authors should preferably report a 2-sigma error bar than state that they have a 96% CI, if the hypothesis of Normality of errors is not verified.
   - For asymmetric distributions, the authors should be careful not to show in tables or figures symmetric error bars that would yield results that are out of range (e.g., negative error rates).
   - If error bars are reported in tables or plots, the authors should explain in the text how they were calculated and reference the corresponding figures or tables in the text.

8. **Experiments compute resources**

   Question: For each experiment, does the paper provide sufficient information on the computer resources (type of compute workers, memory, time of execution) needed to reproduce the experiments?

   Answer: [Yes]

Justification: Appendix F reports the compute worker, wall-clock or exact unit workload, and GPU-hour accounting. It also discloses which response-period launch logs did not retain complete elapsed times instead of imputing totals.

Guidelines:

- The answer [N/A] means that the paper does not include experiments.
- The paper should indicate the type of compute workers CPU or GPU, internal cluster, or cloud provider, including relevant memory and storage.
- The paper should provide the amount of compute required for each of the individual experimental runs as well as estimate the total compute.
- The paper should disclose whether the full research project required more compute than the experiments reported in the paper (e.g., preliminary or failed experiments that didn't make it into the paper).

9. **Code of ethics**

Question: Does the research conducted in the paper conform, in every respect, with the NeurIPS Code of Ethics `https://neurips.cc/public/EthicsGuidelines`?

Answer: [Yes]

Justification: The work is fundamental research on optimization and diagnostics for variational quantum many-body simulation, and it does not involve human subjects, private data, or deployed decision systems.

Guidelines:

- The answer [N/A] means that the authors have not reviewed the NeurIPS Code of Ethics.
- If the authors answer [No], they should explain the special circumstances that require a deviation from the Code of Ethics.
- The authors should make sure to preserve anonymity (e.g., if there is a special consideration due to laws or regulations in their jurisdiction).

10. **Broader impacts**

Question: Does the paper discuss both potential positive societal impacts and negative societal impacts of the work performed?

Answer: [N/A]

Justification: The work studies optimizer regularization for neural quantum states and does not introduce a deployed system or a direct application with identifiable societal impact.

Guidelines:

- The answer [N/A] means that there is no societal impact of the work performed.
- If the authors answer [N/A] or [No], they should explain why their work has no societal impact or why the paper does not address societal impact.
- Examples of negative societal impacts include potential malicious or unintended uses (e.g., disinformation, generating fake profiles, surveillance), fairness considerations (e.g., deployment of technologies that could make decisions that unfairly impact specific groups), privacy considerations, and security considerations.
- The conference expects that many papers will be foundational research and not tied to particular applications, let alone deployments. However, if there is a direct path to any negative applications, the authors should point it out. For example, it is legitimate to point out that an improvement in the quality of generative models could be used to generate Deepfakes for disinformation. On the other hand, it is not needed to point out that a generic algorithm for optimizing neural networks could enable people to train models that generate Deepfakes faster.
- The authors should consider possible harms that could arise when the technology is being used as intended and functioning correctly, harms that could arise when the technology is being used as intended but gives incorrect results, and harms following from (intentional or unintentional) misuse of the technology.

- If there are negative societal impacts, the authors could also discuss possible mitigation strategies (e.g., gated release of models, providing defenses in addition to attacks, mechanisms for monitoring misuse, mechanisms to monitor how a system learns from feedback over time, improving the efficiency and accessibility of ML).

11. **Safeguards**

    Question: Does the paper describe safeguards that have been put in place for responsible release of data or models that have a high risk for misuse (e.g., pre-trained language models, image generators, or scraped datasets)?

    Answer: [N/A]

    Justification: The paper does not release a model, dataset, or system with high misuse risk that would require special safeguards.

    Guidelines:

    - The answer [N/A] means that the paper poses no such risks.
    - Released models that have a high risk for misuse or dual-use should be released with necessary safeguards to allow for controlled use of the model, for example by requiring that users adhere to usage guidelines or restrictions to access the model or implementing safety filters.
    - Datasets that have been scraped from the Internet could pose safety risks. The authors should describe how they avoided releasing unsafe images.
    - We recognize that providing effective safeguards is challenging, and many papers do not require this, but we encourage authors to take this into account and make a best faith effort.

12. **Licenses for existing assets**

    Question: Are the creators or original owners of assets (e.g., code, data, models), used in the paper, properly credited and are the license and terms of use explicitly mentioned and properly respected?

    Answer: [Yes]

    Justification: The experimental tools were developed on top of NetKet (version $\geq$ 3.19), an open-source neural quantum state and variational Monte Carlo library distributed under the Apache License 2.0, which is credited in the Code and Data Availability statement. The paper does not use third-party datasets or pretrained models.

    Guidelines:

    - The answer [N/A] means that the paper does not use existing assets.
    - The authors should cite the original paper that produced the code package or dataset.
    - The authors should state which version of the asset is used and, if possible, include a URL.
    - The name of the license (e.g., CC-BY 4.0) should be included for each asset.
    - For scraped data from a particular source (e.g., website), the copyright and terms of service of that source should be provided.
    - If assets are released, the license, copyright information, and terms of use in the package should be provided. For popular datasets, `paperswithcode.com/datasets` has curated licenses for some datasets. Their licensing guide can help determine the license of a dataset.
    - For existing datasets that are re-packaged, both the original license and the license of the derived asset (if it has changed) should be provided.
    - If this information is not available online, the authors are encouraged to reach out to the asset's creators.

13. **New assets**

    Question: Are new assets introduced in the paper well documented and is the documentation provided alongside the assets?

    Answer: [Yes]

Justification: The paper releases a public repository containing the code and compact artifacts for this work: `https://github.com/takh04/sr_filter`. The repository is linked in Appendix A and includes quickstart and figure-regeneration commands; the paper documents the experimental protocols and compute resources in the appendix.

Guidelines:

- The answer [N/A] means that the paper does not release new assets.
- Researchers should communicate the details of the dataset/code/model as part of their submissions via structured templates. This includes details about training, license, limitations, etc.
- The paper should discuss whether and how consent was obtained from people whose asset is used.
- At submission time, remember to anonymize your assets (if applicable). You can either create an anonymized URL or include an anonymized zip file.

14. **Crowdsourcing and research with human subjects**

Question: For crowdsourcing experiments and research with human subjects, does the paper include the full text of instructions given to participants and screenshots, if applicable, as well as details about compensation (if any)?

Answer: [N/A]

Justification: The work does not involve crowdsourcing or human subjects research.

Guidelines:

- The answer [N/A] means that the paper does not involve crowdsourcing nor research with human subjects.
- Including this information in the supplemental material is fine, but if the main contribution of the paper involves human subjects, then as much detail as possible should be included in the main paper.
- According to the NeurIPS Code of Ethics, workers involved in data collection, curation, or other labor should be paid at least the minimum wage in the country of the data collector.

15. **Institutional review board (IRB) approvals or equivalent for research with human subjects**

Question: Does the paper describe potential risks incurred by study participants, whether such risks were disclosed to the subjects, and whether Institutional Review Board (IRB) approvals (or an equivalent approval/review based on the requirements of your country or institution) were obtained?

Answer: [N/A]

Justification: The work does not involve human subjects, so IRB or equivalent approval is not applicable.

Guidelines:

- The answer [N/A] means that the paper does not involve crowdsourcing nor research with human subjects.
- Depending on the country in which research is conducted, IRB approval (or equivalent) may be required for any human subjects research. If you obtained IRB approval, you should clearly state this in the paper.
- We recognize that the procedures for this may vary significantly between institutions and locations, and we expect authors to adhere to the NeurIPS Code of Ethics and the guidelines for their institution.
- For initial submissions, do not include any information that would break anonymity (if applicable), such as the institution conducting the review.

16. **Declaration of LLM usage**

Question: Does the paper describe the usage of LLMs if it is an important, original, or non-standard component of the core methods in this research? Note that if the LLM is used only for writing, editing, or formatting purposes and does *not* impact the core methodology, scientific rigor, or originality of the research, declaration is not required.

Answer: [N/A]

Justification: LLMs are not part of the scientific method, algorithm, experiments, or evaluation described in the paper.

Guidelines:

- The answer [N/A] means that the core method development in this research does not involve LLMs as any important, original, or non-standard components.
- Please refer to our LLM policy in the NeurIPS handbook for what should or should not be described.